

Project Thesis

Systematic Literature Review on Integration of AI Methods in CFD Simulations

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Abstract (German)

Diese systematische Literaturrecherche zielt darauf ab, den aktuellen Fortschritt und die Forschungslücken bei der Integration künstlicher Intelligenz (KI) in Computational Fluid Dynamics (CFD)-Arbeitsabläufen zu identifizieren. Die Arbeit befasst sich mit zwei Forschungsfragen: 1) Welche Lücken gibt es in der aktuellen Literatur und wie kann die zukünftige Forschung diese Lücken schließen? 2) Inwiefern verbessern KI-Techniken den Designoptimierungsprozess und welche Herausforderungen ergeben sich bei ihrer Anwendung? Aus der SCOPUS-Datenbank wurden nur neuartige, ingenieurwissenschaftlich relevante Ansätze ausgewählt. Die Suche wurde am 23. Mai 2025 durchgeführt, und folgt dem PRISMA-Framework mit qualitativer und quantitativer Analyse. 99 Referenzen aus fünf Bereichen werden befasst: Turbulenzmodellierung (9), Meshing (15), Chemie- und Verbrennungsmodellierung (7), inverse Probleme (14) und Designoptimierung (54). Die Ergebnisse zeigen, dass KI-gestützte CFD ein exploratives Gebiet ist. Surrogatmodelle dominieren die aktuelle Forschung, während physik-informierte neuronale Netze und bestärkendes Lernen an Bedeutung gewinnen. Die Herausforderungen bestehen in der Verallgemeinerbarkeit und der industriellen Anwendbarkeit. Daher sollte die zukünftige Forschung auf breiteren Validierungsdatensätzen und einer tieferen Solver-Integration liegen.

Keywords: numerische Strömungsmechanik, Fluidsimulation, Künstliche Intelligenz, Maschinelles Lernen

Abstract (english)

This systematic literature review aims to evaluate the current progress and identify research gaps in the integration of artificial intelligence (AI) within computational fluid dynamics (CFD) workflows. The review addresses two research questions: 1) What are the gaps in the current literature, and how can future research address these gaps? 2) In what ways do AI techniques enhance the design optimization process within the CFD pipeline, and what challenges arise during their application? Only novel approaches relevant in the engineering domain are chosen from the SCOPUS database, with the search conducted on May 23, 2025. Data synthesis follows the PRISMA framework, combining quantitative and qualitative analyses. In this study, 99 references in five fields are included: turbulence modeling (9), meshing (15), chemical and combustion modeling (7), inverse problems (14), and design optimization (54). Results indicate that AI-assisted CFD remains an exploratory field. Surrogate models dominate current research, while physics-informed neural networks and reinforcement learning are gaining prominence. Key challenges persist in generalizability and industrial applicability, and thus future research should emphasize broader validation datasets and deeper solver integration.

Keywords: computational fluid dynamics, fluid simulation, artificial intelligence, machine learning

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List of abbreviations

AFT	Advancing Front Technique
AI	Artificial Intelligence
AMR	Adaptive Mesh Refinement
ANN	Artificial Neural Network
AWM	Artificial Neural Network-based Wall Stress Model
BMEP	Break Mean Effective Pressure
BO	Bayesian Optimization
BPNN	Backpropagation Neural Networks
CFCFD	Comprehensive Fuel Combustion Flow Dynamics
CFD	Computational Fluid Dynamics
CNN	Convolutional Neural Network
COM	Continuity-Only Model
CSA	Coupled Spalart-Allmaras
DE	Differential Evolution
DeepONet	Deep Operator Network
DGM	Diffusion-based Generative Model
DMD	Dynamic Mode Decomposition
DNN	Deep Neural Network
DNS	Direct Numerical Simulation
DRL	Deep Reinforcement Learning
DRSM	Direct Reynolds Stress Model
ER	Experience Replay
FCFD	Full Combustion Flow Dynamics
FIML	Field Inversion and Machine Learning
GA	Genetic Algorithm
GAN	Generative Adversarial Network
GCN	Graph Convolutional Network
GDNN	Gaussian Deep Neural Networks
GEP	Gene Expression Programming
GIE	Gross Indicated Efficiency
GNN	Graph Neural Network

GP	Gaussian Process
GPR	Gaussian Process Regression
HBNG	Hydrogen-Blended Natural Gas
HFERD	Hybrid Flamelet-Equilibrium Random Data
HPC	High-Performance Computing
LBL	Line-by-line
LES	Large Eddy Simulation
LHS	Latin Hypercube Sampling
LSTM	Long Short-Term Memory
MAE	Mean Absolute Error
MAPE	Mean Absolute Percentage Error
MARL	Multi-Agent Reinforcement Learning
MCMC	Markov Chain Monte Carlo
MLM	Mixing Length Model
MLP	Multi-Layer Perceptron
NN	Neural Network
NRMSE	Normalized Root Mean Squared Error
NSGA-II	Non-Dominated Sorting Algorithm II
PDE	Partial Differential Equation
PINN	Physics-Informed Neural Network
PIRGAN	Physics-Informed Recurrent Generative Adversarial Network
POD	Proper Orthogonal Decomposition
PPO	Proximal Policy Optimization
PSO	Particle Swarm Optimization
RANS	Reynolds-Averaged-Navier-Stokes
RBFNN	Radial Basis Function Neural Network
RCCI	Reactivity Controlled Compression Ignition
RDE	Rotating Detonation Engine
Re	Reynolds number
ReF-ER	Remember and Forget Experience Replay
rel.	relative

RF	Random Forests
RL	Reinforcement Learning
RMSE	Root Mean Squared Error
ROM	Reduced-Order Modeling
RQ	Research Question
RRMSE	Relative Root Mean Squared Error
SA	Spalart-Allmaras
SGS	Subgrid scales
SOM	Self-Organizing Mechanism
SQP	Sequential Quadratic Programming
SR3	Sparse Relaxed Regularized Regression
TBL	Turbulent Boundary Layer
TVKEM	Turbulent Viscosity and Kinetic Energy Model
TVM	Turbulent Viscosity Model
UnTWIST	Unsteady Time-Series Identification of Sparse Transitions
WMLES	Wall-Modeled Large Eddy Simulation
WO	Whale Optimization
WSGG	Weighted Sum of Gray Gases
XGBoost	Extreme Gradient Boosting

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1 Introduction

1.1 Motivation

Computational fluid dynamics (CFD) is a discipline that utilizes computers to generate quantitative predictions of fluid-flow behavior, relying on the fundamental conservation principles of mass, momentum, and energy that govern fluid motion. It is used in many fields to simulate flows on computers, for example in aerospace, mechanical engineering, and process engineering [Hu12]. Although the methods used in calculating these flow properties are now advanced, there is still potential for optimization. Conventional methods reach their limits, especially in turbulent, transient, or multiphase flows. The calculations are computationally intensive and require powerful hardware. Added to this are challenges such as mesh creation, which affects the quality of the results, and complex workflows with manual steps that are error-prone and time-consuming. [Vin22]

In recent years, artificial intelligence (AI) has made great progress and is now used in many technical fields. Within the broader domain of AI, machine learning is concerned with creating algorithms that can extract patterns and make predictions directly from data, thereby avoiding the need for explicitly defined mathematical models. The layered architecture, especially in deep learning applications, enables a hierarchical learning process: lower layers identify simple features in the data, while higher layers integrate these features into increasingly abstract representations. Such a hierarchical structure mirrors the complexity of many physical systems, which often exhibit patterns that can naturally be decomposed into different levels of abstraction. [Vin22]

Also in fluid mechanics, there are initial approaches to using AI to accelerate simulations, enable automated decision-making, predict certain flow patterns and make individual steps of the CFD pipeline such as mesh creation or parameter setting more efficient. [Vin22] Data-driven approaches offers great potential for faster and more accurate simulations but also poses several challenges. [Zha24c] That is why it is essential to analyze current trends and identify gaps, in order to highlight areas where future research can be directed.

1.2 Objective of the work

The main objective of this work is to conduct a systematic literature review to examine the current progress and research gaps in the integration of AI into CFD.

The research questions (RQs) are formulated such that they systematically address both the state of the art and the future potential of integrating AI into CFD workflows. They are designed to find existing gaps in the literature, evaluate the capabilities of current methods, and analyze the challenges that hinder their practical application. The goal is to build a comprehensive understanding of how AI can be leveraged to enhance several steps in the CFD pipeline, with a dedicated focus on design optimization.

RQ1: What are the gaps in the current literature regarding the integration of AI in CFD workflows, and how can future research address these gaps?

This research question aims to critically assess the current state of the literature concerning the integration of AI into CFD workflows. Specifically, it seeks to identify shortcomings, such as methodological limitations, underexplored application domains, or insufficient validation strategies that hinder the systematic adoption of AI-based approaches in CFD. The objective is not only to represent the gaps in knowledge but also to propose potential research directions that could address these gaps and thereby advance the applicability of AI-driven techniques within CFD pipelines.

In order to address this research question, the analysis will be structured around four selected research areas: turbulence modeling, meshing, chemical and combustion modeling, and inverse problems. These domains are chosen because they represent central components of the CFD workflow in which AI has already demonstrated promising potential but where the current state of research remains fragmented and incomplete. Turbulence modeling represents a central area of focus, as it constitutes one of the most computationally demanding aspects of CFD and has long been recognized as a bottleneck for accuracy and efficiency. [Zha24c] Meshing is included because mesh generation and adaptation remain labor-intensive and highly problem-specific tasks, making them a prime candidate for automation through data-driven approaches. [Jai21] Chemical and combustion modeling is selected since reacting flows introduce additional layers of complexity, particularly in multiphysics coupling, where AI methods may help reduce computational cost while maintaining predictive fidelity. [Pau24] Finally, inverse problems are considered because they are inherently ill-posed and computationally intensive, and thus stand to benefit significantly from machine learning techniques that can accelerate optimization and parameter estimation [Zhu22].

RQ2: In what ways do AI techniques enhance the design optimization process within the CFD pipeline, and what challenges arise during their application?

This research question aims to investigate the potential contributions of AI particularly to the design optimization process within CFD workflows, while examining the challenges associated with their implementation. On the one hand, AI techniques offer opportunities to accelerate optimization by reducing computational cost, improving

exploration of high-dimensional design spaces, and enabling the development of surrogate models that approximate expensive CFD simulations. These advantages can significantly enhance the efficiency and scope of design studies, allowing engineers to evaluate a broader range of configurations in less time [Cho22]. On the other hand, the research question also highlights the necessity of addressing the limitations and risks inherent to AI integration. Such challenges include ensuring the reliability and generalizability of surrogate models, managing the large volumes of high-quality training data required, and balancing the trade-off between model accuracy and computational savings [Jal24]. By examining both the benefits and obstacles, this research question seeks to provide a balanced understanding of how AI can reshape design optimization in CFD, and what methodological or practical issues must be overcome for its robust adoption.

1.3 Structure of the work

The structure of this literature review is organized as follows: The current chapter outlines the aim and scope of the review. Chapter 2 presents the state of the art related to the research topic and introduces the fundamental concepts and refining the research questions. It provides an overview of the current conventional practices in several different common steps of the CFD workflow: turbulence modeling, meshing, chemical and combustion modeling, inverse problems and design optimization. Chapter 3 explains the research methodology, including the overall workflow and the qualitative and quantitative evaluation criteria for the systematic literature review. The results of the review are detailed in Chapter 4. Chapter 5 interprets these findings in the context of the defined research objectives and the existing body of literature, and critically discusses their relevance, implications, and limitations. Finally, Chapter 6 concludes the study by summarizing the main insights and proposing potential directions for future research.

2 State of the Art

The integration of AI methods into CFD has gained increasing attention across multiple research domains. This section presents a structured overview of the current state of the art in the five key areas discussed in Chapter 1: Turbulence Modeling, Meshing, Chemical and Combustion Modeling, Inverse Problems, and Design Optimization.

CFD pipeline

CFD is a numerical methodology used to simulate fluid flow, heat transfer, and related physical phenomena by solving the governing equations of fluid motion. To apply CFD in practical engineering problems, a structured workflow is typically followed, which can be divided into four key stages: pre-processing, numerical simulation, post-processing, and validation.

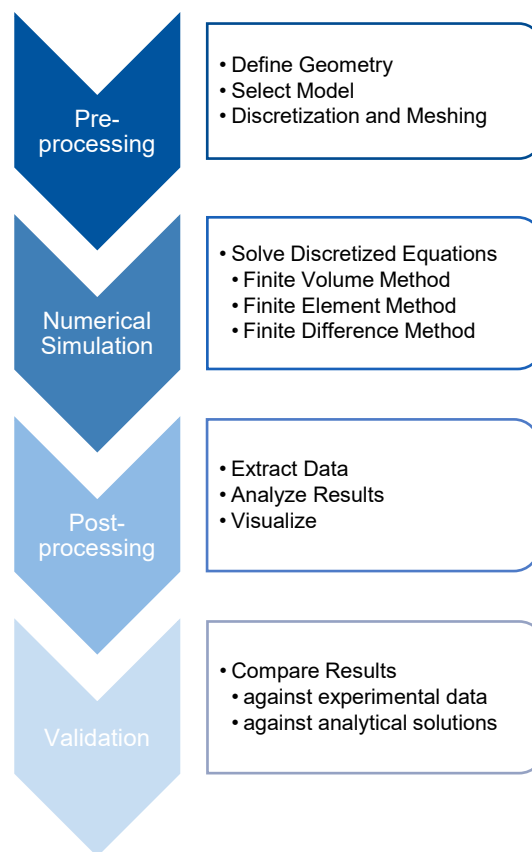


Figure 2.1: CFD Pipeline

In the pre-processing stage, the physical problem is translated into a computational model. This involves defining the geometry of the domain, selecting the relevant physical models (e.g. turbulence models, multiphase flow, heat transfer), and discretizing the domain into a mesh or grid. Meshing refers to the process of discretizing a continuous geometric domain into a finite number of smaller, non-overlapping elements or cells. This spatial discretization enables the numerical solution of the governing equations by approximating them over each element.

Traditional meshing methods have several shortcomings. First, they require manual work and domain knowledge, which hinders automation due to subjective decision-making [Lim24]. This stems from the need to generate meshes tailored to the specific geometry and flow conditions of each case [San24]. The mesh must have high resolution in regions with large gradients of flow properties to capture relevant flow phenomena. Examples include shock waves in transonic and supersonic flows, whose location is hard to predict before running a CFD simulation. Using a uniformly high resolution across the domain is inefficient because it increases computational cost [Yu23]. To balance simulation accuracy and cost, adaptive mesh refinement (AMR) is often used. AMR runs the simulation on an initial suboptimal grid, estimates the error for each element, and improves the mesh locally. This is done iteratively and requires a CFD simulation for each run, making it a time consuming process [Fou23]. In addition to that, traditional mesh generation methods also face convergence issues, especially in complex three-dimensional geometries, again requiring human intervention [Liu24c]. For these reasons, meshing remains one of the bottlenecks in CFD simulations [Loc23].

The numerical simulation stage involves solving the discretized form of the governing equations. Depending on the solver, this may be done using finite volume, finite element, or finite difference methods. The solution process typically requires significant computational resources, especially for three-dimensional, transient, or turbulent flows.

In the post-processing phase, the simulation results are analyzed and visualized. This includes extracting quantities of interest such as pressure distributions, velocity profiles, or aerodynamic coefficients. Visualization tools help in understanding flow structures like vortices, shock waves, or boundary layer separation [Pan23b].

Finally, validation is carried out by comparing simulation results against experimental data or analytical solutions. This step is essential to ensure the reliability of the numerical predictions and to identify possible limitations or modeling errors in the setup [HIR06].

This established workflow forms the basis for integrating AI-based methods, which aim to enhance or replace parts of the CFD process to reduce computational cost, increase automation or to improve the quality of the outcome of one of the listed steps above.

Different turbulence modeling approaches

In real-world scenarios, including most industrial applications, the majority of flow problems are turbulent and therefore inherently multi-scale. Multi-scale means that turbulence generates a wide range of interacting length and time scales within the flow. In the same flow field, there can be large swirls or vortices, called “eddies”, (representing the large scales), alongside tiny, chaotic fluctuations (representing the small scales), where kinetic energy is ultimately converted into heat through viscous dissipation. The energy is initially introduced at large scales and then transferred progressively to smaller scales through a process known as the “energy cascade”, until the energy is

viscously dissipated. This cascade, introduced by LEWIS FRY RICHARDSON in 1922 [Ric22] and formalized by ANDREY KOLMOGOROV in 1941 [Kol41] illustrates the complexity of turbulent flows, where different sizes and timescales continuously interact with each other. [Pop00; Kol41]

To address the multi-scale nature of complex turbulent flows, three primary computational approaches are commonly employed: Direct Numerical Simulation (DNS), Reynolds-Averaged Navier–Stokes (RANS), and Large Eddy Simulation (LES). Each method follows a distinct philosophy for resolving different flow scales and reflects different trade-offs between computational cost and physical accuracy.

DNS is the most accurate but also the most computationally demanding approach. It solves the full Navier–Stokes equations without any approximations or modeling, explicitly resolving all scales in the energy cascade, from the largest eddies down to the smallest [Col10]. In DNS, the computational grids are selected to be sufficiently fine to resolve the Kolmogorov length scale,

$$\eta = \left(\frac{\nu^3}{\varepsilon} \right)^{1/4}, \quad (2.1)$$

where ν denotes the kinematic viscosity and ε represents the dissipation rate. Kolmogorov scales characterize the smallest dissipative eddies. [Pop00; Col10] For homogeneous isotropic turbulence, the total number of required grid points scales as $Re_T^{9/4}$, with Re_T being the turbulence Reynolds number [Col10]. This results in extremely high computational costs, making DNS practically intractable for industrial applications, where turbulent flows dominate, and thus Reynolds numbers are typically large.

RANS, based on the Reynolds-averaged Navier–Stokes equations introduced by OSBORNE REYNOLDS in 1895 [Rey94], is the most computationally efficient modeling method. Reynolds' foundational work decomposed flow variables into mean and fluctuating components. By applying time-averaging to the Navier–Stokes equations, the RANS formulation introduces additional terms known as Reynolds stresses $-\rho \overline{u_i u_j}$, with u_i and u_j the velocity components and ρ the density, which capture the effects of turbulent fluctuations on the mean flow. RANS methods provide reliable and inexpensive predictions of mean flow properties for complex geometries at high Reynolds numbers, although they have limitations in capturing unsteady flow phenomena and turbulent fluctuations. For these reasons, RANS remains the primary tool for industrial CFD applications. [Sha04]

LES represents a middle ground between DNS and RANS by resolving the large-scale turbulent motions while modeling the smaller scales. The concept of LES was first proposed in the early 1960s by JOSEPH SMAGORINSKY [Sma63]. The method applies spatial filtering to separate resolved scales from subgrid scales (SGS), relying on the principle that large turbulent structures depend on the geometry, whereas smaller scales exhibit more universal behavior, making them suitable for modeling. The universality of the smaller scales is demonstrated by KOLMOGOROV's hypothesis of local isotropy:

“At sufficiently high Reynolds number, the small-scale turbulent motions are statistically isotropic” [Pop00; Kol41]

LES captures the time-dependent behavior of resolved scales, while RANS provides time-averaged statistics. LES scales more favorably with Reynolds number than DNS, but its computational cost remains significantly higher than that of RANS. [Pio14; Pop00]

In Figure 2.2, the energy cascade is depicted, with $\log(k)$ logarithm of the wave number and $\log(E(k))$ logarithm of the energy, dependent on the wave number; as well as which areas of the energy cascade is resolved or modeled for DNS, LES and RANS.

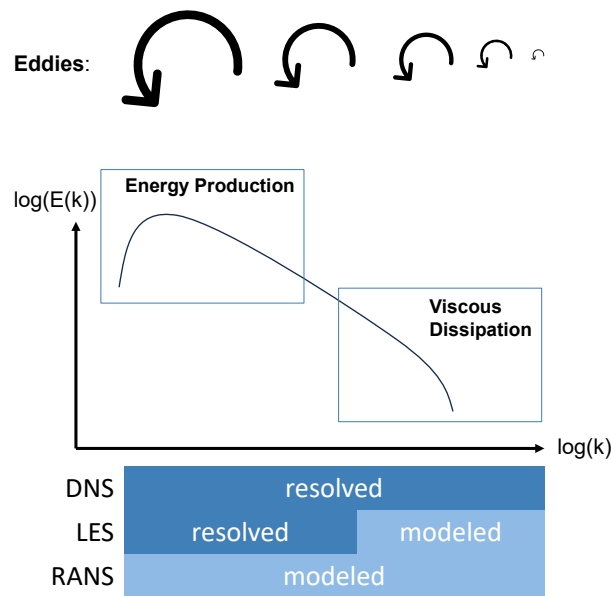


Figure 2.2: Energy cascade and DNS/LES/RANS

However, simplifying the governing equations also imposes an additional problem that needs to be addressed: the closure problem. In RANS, this arises because the symmetric Reynolds stress tensor $-\rho \overline{u_i u_j}$ introduces six additional unknowns in three-dimensional flows, resulting in more unknowns than available equations [Spe91]. RANS turbulence models resolve this issue using different approaches: Zero-equation (algebraic) models, which relate turbulent viscosity directly to other flow variables; one-equation models, which involve solving a transport equation for one turbulent quantity; and two-equation models, which require solving transport equations for two turbulent quantities [Spe91]. In LES, the filtered Navier–Stokes equations contain subgrid-scale (SGS) stress terms that must be modeled. These SGS stresses account for the influence of unresolved small-scale motions on the resolved large-scale flow structures. Similar to RANS, this leads to a closure problem; however, as discussed before, it relies on universal small-scale physics [Men00; Pop00]. One of the pioneering approaches to address this is the Smagorinsky model, proposed by JOSEPH SMAGORINSKY [Sma63].

Common AI approaches

Artificial Intelligence (AI) constitutes a broad class of computational methodologies that facilitate the extraction of meaningful patterns, robust predictions, and decision-making strategies from complex, high-dimensional datasets. High-fidelity CFD, like large-eddy or direct numerical Simulation, offers accuracy in simulating complex flows but at a high computational cost, often requiring extensive HPC (High-Performance Computing) resources and long runtimes that make routine design iterations impractical. In response, modern AI techniques are applied within the simulation pipeline to alleviate these computational burdens and extend CFD capabilities. AI Methods used in the CFD pipeline can be categorized into five primary groups: Neural Networks, Surrogate or Statistical Models, Physics-Informed and Hybrid Models, Reinforcement Learning. [Zha21; Zha25a; Bru21b]

Neural networks are data-driven models composed of interconnected neurons arranged in layered architectures. Key variants include Artificial Neural Networks (ANNs), Deep Neural Networks (DNNs), networks characterized by multiple hidden layers, and Convolutional Neural Networks (CNNs), specialized for processing structured, grid-like data [Zha24c; Obi23]. Neural network architectures have also evolved to handle more irregular data formats effectively. Specifically, Graph Neural Networks (GNNs) and Graph Convolutional Networks (GCNs) are specialized neural networks designed explicitly for graph-structured data. These graph-based architectures leverage node-to-node relationships and the inherent topology within data, enabling learning directly on graphs without requiring transformation into structured grids [Pel24]. A further subclass is Generative AI models, such as Generative Adversarial Networks (GANs) and diffusion models, which are designed to learn data distributions and produce new samples with similar statistical properties [Wan24c]. Bayesian diffusion models combine generative diffusion processes with probabilistic inference, combining sample generation with uncertainty quantification. Long Short-Term Memory (LSTM) networks are another type of NNs designed for sequential or time-dependent data. They use memory cells and gating mechanisms, like input output and forget gates, to regulate information flow enabling the capture of both short and long-term data dependencies [Fan25].

Surrogate models provide computationally efficient approximations of more complex or computationally intensive models. Gaussian Process Regression (GPR) is a statistical modeling approach that offers predictions along with uncertainty quantification [Taj23]. Reduced-order modeling (ROM) methods, such as Proper Orthogonal Decomposition (POD) and Dynamic Mode Decomposition (DMD), represent complex systems using simplified, lower-dimensional formulations. ROMs are often integrated with machine learning algorithms to enhance predictive capabilities [Men21].

Physics-Informed Neural Networks (PINNs) integrate physical laws directly within their learning process, typically embedding governing equations into their training objectives or loss functions. Hybrid models combine traditional numerical solvers with data-driven

components, such as neural networks or surrogate models, to leverage the strengths of each methodology [Rai19].

Reinforcement learning (RL) is an area of machine learning where an agent learns optimal sequences of actions and observes the resulting states by interacting with an environment. The agent receives rewards that quantify the outcome of its action, and the objective is to maximize the cumulative reward over time. The mapping from states to actions is defined by a policy, which is iteratively improved through exploration and feedback [Pan23a]. Deep Reinforcement Learning (DRL) extends RL by incorporating neural networks to approximate value functions or policies, enabling the agent to learn complex decision-making processes from high-dimensional inputs [Wan24a].

Supervised learning refers to machine learning methods trained on labeled datasets, where each input is paired with a known output. The objective is to approximate a mapping function that predicts outputs for unseen inputs. Unsupervised learning refers methods trained on unlabeled datasets, where the objective is to identify hidden structures or patterns within the data [Yu23].

While supervised and unsupervised learning are trained on datasets provided beforehand, reinforcement learning differs from both by framing the problem as an interaction between an agent and an environment. [Nov20]

2.1 Turbulence Modeling

As outlined previously, the objective of turbulence modeling is to approximate the influence of the unresolved fluctuations from the RANS or LES approximations, and thus “closing” the equation systems. These fluctuations have an influence on the mean flow quantities of engineering relevance such as lift, drag, or the location of flow separation over aerodynamic surfaces like airfoils. Therefore, turbulence closure requires developing reliable models that can effectively represent the net effect of turbulence on averaged flow behavior without resolving the full complexity of the underlying flow structures [Bru21a].

However, many conventional turbulence closure models such as RANS-based eddy-viscosity formulations (k - ϵ , k - ω , Spalart–Allmaras) and SGS LES models rely on the Boussinesq hypothesis, postulated by JOSEPH BOUSSINESQ in 1877 [Bou77] which assumes a linear relation between Reynolds stresses and mean strain rates, as well as that the turbulence is isotropic. [Sch07] While efficient, these models suffer from universality limitations, and difficulties in complex or anisotropic flows [Zha24c]. Especially anisotropic turbulence and separated flows pose significant problems for conventional RANS and LES closure models [Rum22]. This leads to the RANS and LES closure models to not be generalizable to a significant portion of complex flow situations [Fan25; Zha24c].

In order to address these shortcomings, data-driven approaches like machine learning are researched to modify turbulence models. According to ZHANG ET AL. (2025) [Zha25a], AI methods applied to turbulence modeling in fluid mechanics are categorized under two principal sub-directions: dimensionality reduction/knowledge discovery

and machine learning-based turbulence modeling. The former leverages methods that automatically identify interpretable governing equations or physical laws for turbulent flows from high dimensional data. In contrast, the latter focuses on replacing or augmenting traditional closure models, such as those used in RANS or LES, with data-driven alternatives. For example, this includes deep neural networks trained on high-fidelity DNS or LES data and PINNs that incorporate physical constraints.

The first approaches of utilizing ML and applied statistical inference for turbulence modeling has the goal of calibrating the uncertainties in the RANS model and is pioneered by ZHANG & DURASAMY [Zha15]. This approach is subsequently refined by PARISH & DURASAMY [Par16], who introduces a field-inversion and machine learning (FIML) paradigm that embeds inversion-based discrepancy extraction into traditional solvers. The FIML framework is a method designed to improve turbulence models by correcting their systematic errors (discrepancies) through data-driven adjustments. The problem is formulated as an inverse problem: instead of running the model purely in a forward manner, field values such as source terms are modified so that the simulation output aligns more closely with available data. Machine learning is then employed to capture and learn the patterns of these discrepancies. Once trained, the machine learning model can be integrated into the simulator to provide discrepancy corrections automatically. This eliminates the need for additional high-fidelity data. YAN ET AL. applies field inversion and ML to correct the Spalart–Allmaras model and demonstrates generalization to flow scenarios similar to those of the training dataset [Yan22]. The same year as PARISH & DURASAMY, J. LING ET AL. [Lin16] pioneers the use of deep neural networks in RANS turbulence modeling by building on the work of S. B. POPE from 1975 [Pop75]. S. B. POPE proposed ten basis tensors that can be combined to formulate the Reynolds stresses. This poses an architecture that is based on physical argumentation as well as provides a guide to build neural networks to model these tensors in a way to solve the turbulence closure problem.

One key advantage of using AI for flow predictions is that it enables the development of corrections based on large datasets, which facilitates their application to a wide variety of flow scenarios and allows for potential generalization to flow conditions resembling those found in the training data. Consequently, data-driven methods hold significant promise for enhancing the accuracy of turbulence closure. [Zha24c] In addition to the generalizability aspect, such methods also benefit from significant reductions in runtime once trained [Fox24].

2.2 Meshing

The meshing process in CFD simulations can be divided into two main categories. The first is the generation of a mesh from a given geometry while the second is the improvement of an already generated mesh.

Artificial neural networks (ANNs) are used to address shortcomings in traditional mesh creation algorithms, such as the advancing front technique (AFT), which generates a grid layer by layer, starting from the boundary of the computational domain. LIU ET AL. integrate ANNs into the 3D AFT to accelerate the mesh generation, improve the mesh

quality and overcome convergence issues [Liu24c]. A neural network (NN) that is trained on accurate CFD solutions to predict a near-optimal mesh for unseen geometries is proposed by LOCK ET AL. Their approach uses geometry and flow parameters as input to generate a mesh that incorporates predicted flow features, such as shock waves [Loc23]. A third method, proposed by LIM ET AL., leverages Deep Reinforcement Learning (DRL) to construct meshes. In this approach, the DRL agent incrementally builds the mesh layer by layer from a surface, guided by a reward function based on mesh quality metrics [Lim24].

The category of mesh improvement can be further divided into the two sub-categories smoothing and adaptation. Mesh smoothing is a post-processing technique used to improve mesh quality by adjusting the positions of nodes without altering the mesh topology. The objective is to enhance element shape and distribution. Traditionally either heuristic or optimization-based methods are used for mesh smoothing. Heuristic methods are simpler, but do not necessarily provide the optimal solution. In comparison optimization based methods are slower while offering the optimal solution [Wan25a]. HO ET AL. introduce a method based on Graph Convolutional Neural Networks (GCNs) that optimizes node positions without requiring training data, producing higher-quality meshes than conventional methods [Ho24]. While not faster, this approach offers superior quality. In contrast, WANG ET AL. employ Graph Neural Networks (GNNs) to smooth meshes over ten times faster than optimization-based methods, albeit with comparable mesh quality [Wan25a]. A method that is both faster and provides a higher quality mesh employs DRL to move nodes and optimize a reward function based on mesh metrics [Wan24a].

Mesh adaptation refers to the modification of the mesh based on flow features. Traditionally, adaptation is performed iteratively for steady-state problems, requiring multiple simulations after each mesh modification guided by error estimation [Obi23]. WU ET AL. eliminate these additional simulations by using an NN to predict the flow after running only the initial CFD simulation. The mesh density is then increased iteratively by moving nodes to regions with a large predicted gradient without creating new nodes [WU22]. Similar methods use GNNs [Pel24] or a combination of a Multilayer Perceptron (MLP) and a self-organizing mechanism (SOM) to reposition nodes [Yu23]. To further enhance CFD simulation efficiency, FOUCART ET AL. formulate mesh adaptation as a DRL problem. Their agent is guided by the predicted flow field and a reward function that balances the trade-off between improved accuracy and increased computational cost, thereby determining whether to locally refine or coarsen the mesh [Fou23].

2.3 Chemical and Combustion Modeling

Recent advances integrate AI techniques with CFD simulations to address persistent challenges in chemical and combustion modeling, including the high computational costs of solving detailed chemical kinetics [Wan25c], limited availability of experimental data for validation [Jai21], and challenges in coupling turbulence, radiation, and multi-species chemistry [Men21]. These limitations motivate the introduction of AI methods

to accelerate simulations, develop accurate surrogate models, and support optimization under complex operating conditions [Xue23].

Research focuses on diverse combustion systems such as coal-fired boilers, reactivity-controlled compression ignition (RCCI) engines, rotating detonation engines (RDEs), and supercritical combustion setups. These systems involve multi-scale, multi-physics phenomena that benefit from AI-based surrogate modeling and data-driven methods. [Wan25c; Jai21; Men21; Cai25]

Machine learning (ML) and reduced-order models (ROMs) show strong performance in high-dimensional combustion simulations. For example, in coal-fired boilers, Proper Orthogonal Decomposition combined with Extreme Gradient Boosting (POD-XGBoost[Wan25c] and Gaussian Deep Neural Networks [Xue23] achieves median normalized root mean squared errors (NRMSE) below 4% for temperature, O₂, and CO fields. These approaches require less than 10% of the computational time of traditional CFD simulations, enabling spatially resolved predictions suitable for real-time monitoring and control.

AI is also utilized for optimization of combustion under emission constraints. In RCCI engines, artificial neural networks (ANNs) are trained on CFD data and coupled with genetic algorithms (GA) to optimize parameters such as injection timing and intake charge temperature. This method leads to a 19% increase in indicated thermal efficiency, along with 60% and 64% reductions in hydrocarbon and CO emissions, respectively, compared to baseline cases [Jai21].

For RDEs, where combustion wave dynamics involve strong translational invariance, traditional ROMs face limitations. Deep learning strategies, such as dynamic mode decomposition and deep Koopman embeddings, are applied to identify moving coordinate frames and reduce model errors. These methods enable stable and low-dimensional representations of complex detonation wave behavior, suitable for long-time integration and parametric studies across engine geometries [Men21].

Radiative heat transfer modeling is also improved through AI. A physics-informed neural network (PINN)-based Weighted Sum of Gray Gases (WSGG) model is developed to predict emissivity and absorption coefficients for multi-species gas mixtures. The WSGG model is a simplified method for modeling gas radiation, which approximates the complex spectral behavior of real gases as a sum of several gray gases, each with constant absorption properties, weighted to match the overall spectral characteristics. Compared to traditional curve-fitting approaches, this model offers better accuracy and physical consistency [Che25a]

Chemical kinetics and thermodynamic property modeling under supercritical conditions are addressed using multi-layer perceptron (MLP) surrogates trained on hybrid flamelet-equilibrium random data (HFERD). This approach reduces computational time for chemistry and thermodynamics by 90–96%, with overall simulation speedups between 3× and 5× [Cai25].

In safety-related applications, such as hydrogen-blended natural gas (HBNG) pipelines, ML models are trained on CFD-generated jet fire data. These models predict fire characteristics such as length, temperature, and radiative heat flux more accurately than empirical correlations. The surrogate models also expand the applicable parameter range and support rapid consequence analysis for risk assessment [Zha25b].

These studies collectively highlight the increasing maturity of AI-CFD integration. The fusion of physics-based solvers with data-driven surrogates enables high-resolution, low-latency, and physically consistent modeling across diverse combustion regimes.

2.4 Inverse Problems

Inverse problems in CFD are challenging due to their ill-posedness, meaning that solutions may be non-unique, unstable, or sensitive to noise in the observed data. Traditional techniques often require regularization, rely on repeated forward simulations, and demand high computational cost, particularly when dealing with high-dimensional parameter spaces or multiphysics interactions [Zhu22]. These limitations hinder their applicability in real-time or design-critical settings. Moreover, data availability is often sparse or partial, especially in experimental environments. As a result, there is a growing need for methods that can handle incomplete data, reduce computational burden, and maintain physical consistency [Bru22].

AI surrogates have been deployed in aerodynamic inverse design to replace iterative adjoint-based solvers. In transonic blade optimization, generative neural networks and multi-output regression models have been trained to map desired Mach number distributions to blade geometries [Jia]. These models eliminate the need for costly forward simulations in each design iteration and can generate viable geometries under multiple conditions. This approach has enabled the reconstruction of turbomachinery blades within acceptable aerodynamic performance envelopes while reducing the design cycle time significantly [Yan25].

In thermal systems, inverse estimation of internal heat sources and boundary fluxes has been enhanced through neural architectures that embed the governing physics [Zhu22]. Physics-informed neural networks (PINNs) have been applied to recover spatially distributed heat sources in conjugate solid-fluid domains by incorporating the heat equation directly into the loss function. These models learn from sparse temperature measurements and produce stable inverse estimates even under partial observability [Jal25]. In high-fidelity conjugate simulations, PINNs have demonstrated accurate recovery of heat flux profiles at a fraction of the computational cost required by classical regularized solvers [Cai24].

Inverse modeling in multiphase flows presents additional complexity due to interface dynamics and discontinuities. In recent studies, PINNs have been successfully applied to simulate and recover the behavior of rising bubbles in two-phase flows [Rad24]. These models are able to reconstruct interface motion and flow patterns with high fidelity, even in cases where the velocity field was entirely excluded from the input [Jal25]. The predictive capability of the model extends to fluid systems with varying

material properties and flow regimes, demonstrating generalization across different Reynolds and Bond number conditions [Cui25].

To reduce simulation cost and enable parametric inversion, surrogate-assisted frameworks have been introduced [Zhu22]. The SECRET framework (Statistical Emulation for Computational Reverse Engineering Tasks) employs Gaussian process emulators and Bayesian model selection to perform inverse reconstruction of flow and pressure fields in vascular systems. While originally applied in biomedical domains, its structure generalizes to mechanical and thermal systems. SECRET achieves accurate parameter recovery with uncertainty quantification while using fewer than 5% of the forward simulations required by traditional optimization pipelines [Pau24].

These methods demonstrate the growing maturity of AI-enabled inverse modeling. From PINNs to statistical emulators and geometry-to-flow regressors, such frameworks allow for rapid, interpretable, and constraint-aware reconstruction of system inputs.

2.5 Design Optimization

Design optimization is the focus of the second research question of this literature review. Today, the standard approach to design optimization in CFD often involves coupling the simulation with random or heuristic search methods. For example, heuristic search methods such as genetic algorithms have long been applied to explore CFD design parameter spaces, in order to find the optimal parameters that are identified to lead to an optimal design [Li22]. Using a heuristic approach or brute-force approaches like random search typically requires running thousands of CFD simulations to explore the design space, which leads to prohibitively high computational costs, especially in high-dimensional or complex design problems [Naz25]. Even if a single simulation takes minutes, exploring thousands of designs can take days or weeks [Naz25].

To address this challenge, surrogate modeling, a machine learning approach, gains significant attention. Surrogate models serve as efficient approximations of the expensive CFD solver by learning a mapping from input design parameters to output performance indicators, such as lift-to-drag ratio [Nik25]. Once trained, these models serve as an efficient approximation of the expensive CFD solver and allow rapid evaluations of new designs with acceptable accuracy and without the need for repeated full simulations [Nik25]. This efficiency enables the integration of advanced AI-driven search strategies such as genetic algorithms and Bayesian optimization to explore the design space more effectively [Jia25; Xu24a]. In this context, surrogate models not only accelerate the optimization process but also facilitate broader investigations of trade-offs and uncertainties [Kea20]. Because surrogates execute quickly, it is possible to evaluate many designs to map out Pareto-optimal trade-offs among design objectives, like lift vs. drag, and to perform uncertainty quantification or robust design studies [Mar23; Kea20]. Pareto-optimal trade-offs, first introduced by FRANCIS YSIDRO EDGEWORTH in 1881 and generalized by VILFREDO PARETO in 1896, are central in design optimization using surrogate modeling approaches. They refer to solution in multi-objective optimization, where improving one objective (like reducing drag) cannot happen without

worsening another (like reducing lift). The set of such optimal compromises forms a “Pareto front”, which helps explore balanced design choices in CFD [Coe06]. An example of this for two objectives is shown in Figure 2.3.

The most common workflow for surrogate-based design optimization begins with a set of CFD simulations. In each simulation, the design parameters are varied within pre-defined ranges. The resulting CFD data must sufficiently cover the design space, as it serves as the training set for the surrogate model. Once trained, the model predicts

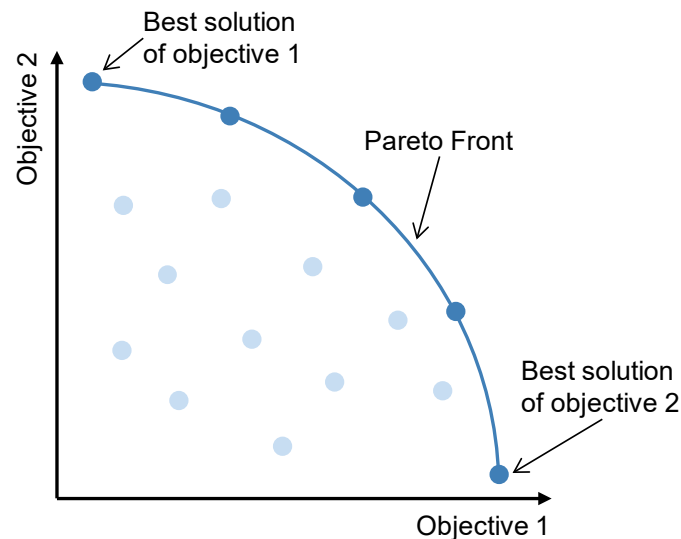


Figure 2.3: Example of a Pareto front

key performance metrics, such as the lift-to-drag ratio, for unseen geometric parameter sets. Heuristic optimization algorithms, such as genetic algorithms or particle swarm optimization, are then applied to identify the parameter set that maximizes or minimizes the target function [Du25].

In both academic and industrial applications, RANS simulations remain practical and widely used in design optimization than LES. The primary reason is that LES methods are prohibitively expensive for the many simulations needed in an optimization loop [Mat25]. Only a handful of recent works attempt LES-based shape optimizations that are limited to low Reynolds-number cases. These are considered proof-of-concept because such approaches do not scale to the high Reynolds numbers of practical engineering flows [Mat25]. Consequently, the engineering community still largely relies on RANS for design, despite RANS turbulence models’ known limitations [Gor22]. Therefore, most optimized designs today are effectively optimal under RANS assumptions, which is acceptable for tractability, but can misrepresent the true performance [Mat25].

3 Methodology: Systematic Literature Review

This chapter provides a detailed description of the methodology of the systematic literature review employed in the study to ensure reproducibility. The chapter is structured into four main parts: the overall research approach, the outline of the framework used for the systematic literature review, and the procedures for both the qualitative and quantitative analysis.

3.1 Research Approach and Search Strategy

To ensure a systematic and rigorous approach to the literature review, this study adopts the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) framework, as stated by PAGE ET. AL. in the PRISMA 2020 statement paper [Pag21]. PRISMA is a widely recognized methodology designed to enhance the quality, transparency, and reproducibility of systematic reviews and to standardize them.

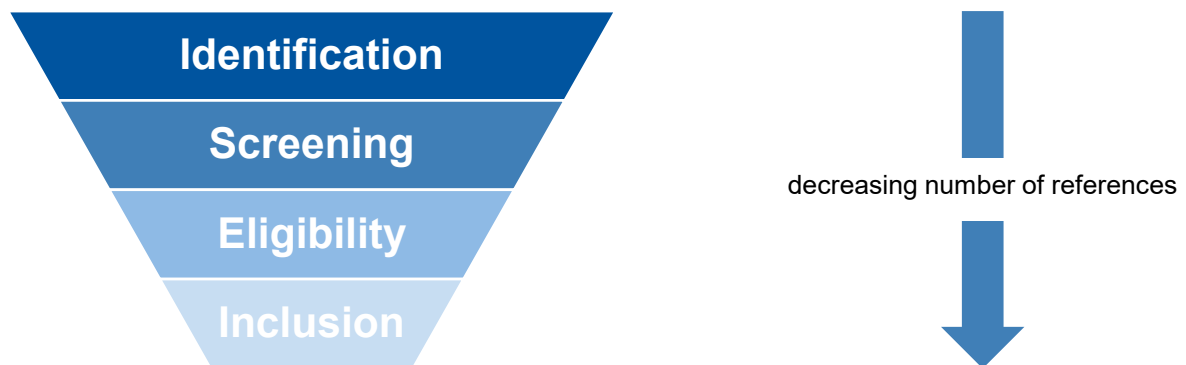


Figure 3.1: Steps of the PRISMA Framework

Following the last step in the PRISMA framework, a set of evaluation criteria is developed to support the qualitative analysis. These criteria are directly linked to the research questions and serve as metrics for assessing the extent to which each study addresses specific challenges, or reveals gaps in the literature. They also provide a consistent perspective for reading and interpreting the selected papers.

3.2 Literature Search with PRISMA

As seen in Figure 3.2, the PRISMA flow diagram is a standardized tool used to transparently document the study selection process. It visually outlines the steps taken from the initial retrieval of records to the final inclusion of studies. The diagram consists of four main phases: Identification, Screening, Eligibility (here, part of Screening), and Inclusion, which will be explained in the following chapters.

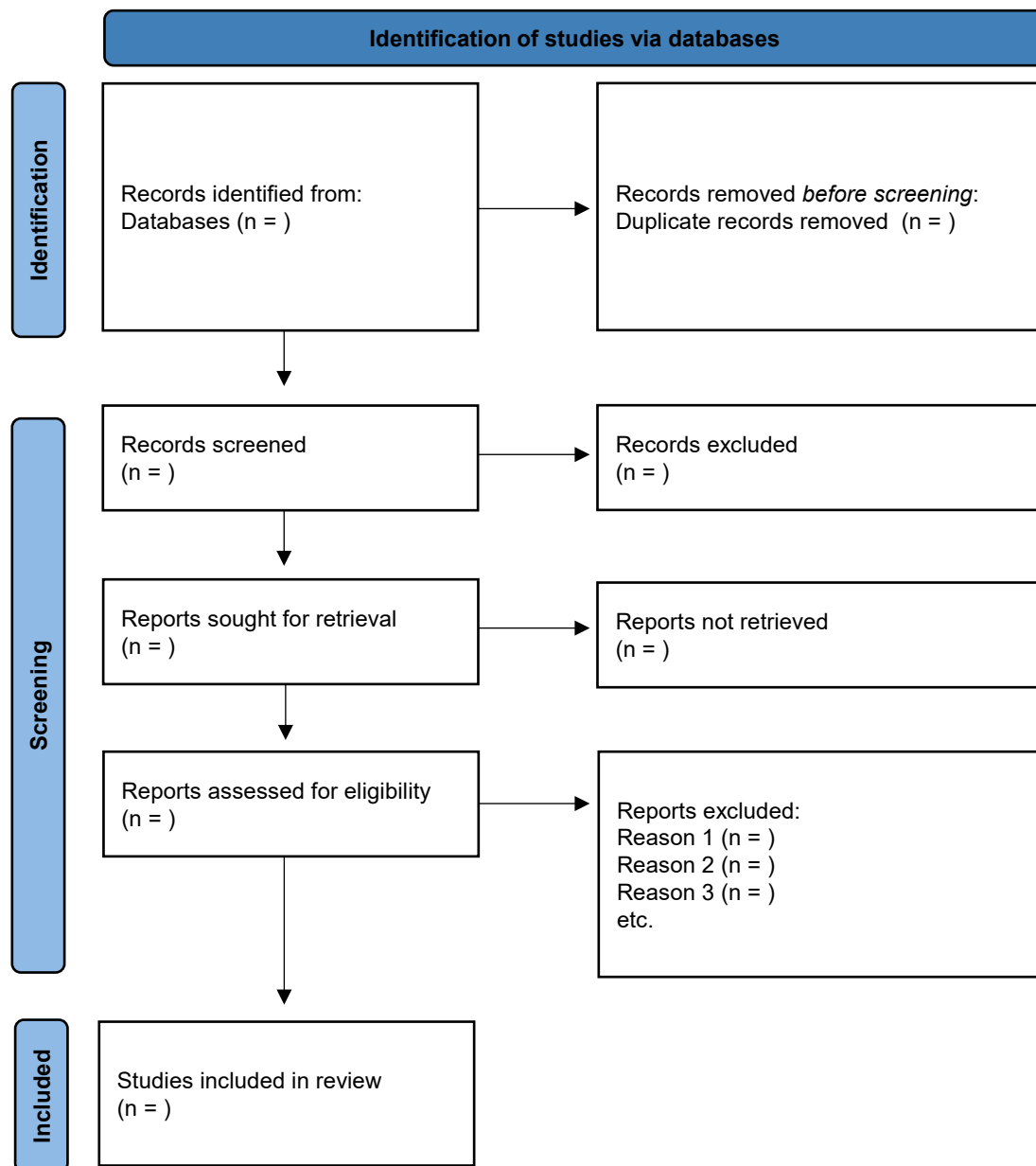


Figure 3.2: Empty PRISMA Flow Diagram (modified) [Pag21]

3.2.1 Identification

The Identification phase begins with the formulation of a search string that is developed based on combinations of keywords relevant to the research topic. This search is applied to the databases that are chosen to be appropriate for the subject domain. This selection is based on its relevance, coverage of peer-reviewed literature, and compatibility with the field of CFD, AI and engineering in general. Once the initial results are retrieved, duplicate entries across databases are removed to prevent overrepresentation of any individual study.

For this study, Scopus is selected as the database due to its broad interdisciplinary coverage, high-quality indexing of peer-reviewed literature, and Boolean search capabilities. Choosing a single, comprehensive database helps mitigate retrieval bias by ensuring consistency and reproducibility across the search process. Relying on

manual or individual selection of papers can introduce selection bias, whereas a standardized database-driven search reduces subjectivity and enhances transparency.

The initial search query is constructed to broadly capture literature at the intersection of CFD and AI, without limiting the results to any specific integration area. The query is implemented using the Boolean search syntax (AND, OR, NOT) of Scopus, which allows field-specific keyword targeting through the TITLE-ABS-KEY command. This ensures that all retrieved documents include the specified terms in either the title, abstract, or keywords.

```
( TITLE-ABS-KEY ( "computational fluid dynamics" OR "cfd" OR "flow simulation*" OR
"liquid simulation*" OR "gas simulation*" OR "fluid simulation*" OR "numerical fluid
mechanics" OR "fluid flow computation" OR "computational flow analysis" OR "simulation
of fluid dynamics" OR "flow model*" OR "fluid dynamics simulation*" ) )

AND ( TITLE-ABS-KEY ( "machine learning" OR "artificial intelligence" OR "deep
learning" OR "deep neural network*" OR "reinforcement learning" OR "deep reinforcement
learning" OR "physics informed neural network*" OR "physics informed machine learning"
OR "Bayesian optimization" OR "transfer learning" OR "generative AI" OR "convolutional
neural network*" OR "recurrent neural network*" OR pinn* OR metamodel* OR "data-based
model*" OR "neural network*" OR "surrogate model*" OR "pattern recogn*" OR "predictive
model*" OR "parameter tun*" OR "reduced order model*" ) )

AND ( LIMIT-TO ( SRCTYPE , "j" ) )

AND ( LIMIT-TO ( SUBJAREA , "ENGI" ) OR LIMIT-TO ( SUBJAREA , "COMP" ) OR LIMIT-TO (
SUBJAREA , "MATH" ) )

AND ( LIMIT-TO ( LANGUAGE , "English" ) )
```

Figure 3.3: The initial search query

The first part of the query targets the CFD domain. The use of wildcards (e.g., *) helps capture variations in word endings (e.g., "model" and "models"). The second part of the query focuses on AI-related terms. To ensure quality and relevance, the query is restricted to journal articles (LIMIT-TO (SRCTYPE , "j")) published in English, and within subject areas engineering, computer science, or mathematics.

Following the initial query, a total of **4,555** papers are retrieved from the Scopus database (retrieval date: 23.05.2025). At this stage, no restrictions are imposed regarding specific application domains within CFD and AI. Given that the second research question in this study focuses specifically on design optimization, the scope of the literature included for the first research question is narrowed to avoid thematic overlap and ensure a more balanced analysis across domains.

As a result, four application areas, as discussed in Chapter 2, are selected for focused investigation of the first research question: turbulence modeling, meshing, chemical and combustion modeling, and inverse problems. These areas are chosen based on their fundamental role in CFD workflows and the challenges they pose. Turbulence modeling, for example, remains a central challenge in CFD due to the closure problem and model-form uncertainty; meshing is a time-intensive and error-prone step in the simulation pipeline; chemical and combustion modeling is critical for reacting flows and combustion simulations; and inverse problems represent a class of applications where AI techniques can reconstruct inputs from observable flow behavior.

To ensure the relevance and recency of the selected literature, cut-off years are defined individually for each of the five research areas. This decision is guided by a temporal distribution analysis of the retrieved publications. First, a graph showing the annual publication count for the entire dataset (Figure 3.4), reveals a notable increase in research activity related to AI integration in CFD over the past decade. Subsequently, separate trend analyzes are performed for each of the four main research areas, as seen in the Figure 3.5. These graphs clearly show that the breakout years, or points of significant publication growth, varied across domains. Therefore, the followings years are chosen:

Table 3.1: Cut-off dates for the different research areas

Research Area	Cut-off Date
Turbulence Modeling	2019
Meshing	2018
Chemical and Combustion Modeling	2018
Inverse Problems	2017
Design Optimization	2022

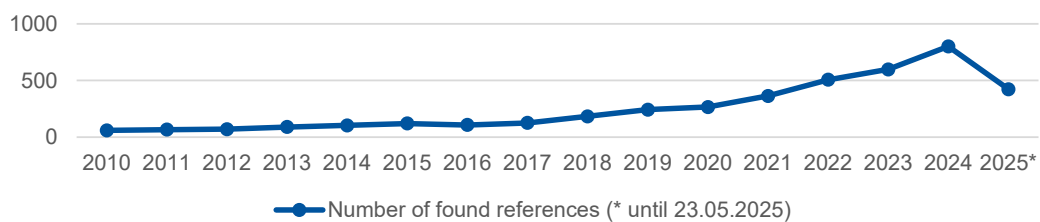


Figure 3.4: Number of found references (without distinction of research area) over the years with the general search string

To address RQ1 more directly, the original search string is refined to include terms related to research gaps, limitations, and future work. This ensures that the retrieved literature explicitly discusses unresolved challenges, barriers to adoption, and possible directions for further investigation. This targeted refinement helps to systematically capture the state of the art while highlighting fields for further research, without producing an unmanageably large or thematically scattered dataset. This restriction is not implemented for addressing meshing and RQ2. For meshing, when these constraints are applied, only four papers meet the inclusion criteria. However, these publications do not adequately represent the current state of the art of AI integration in meshing. Consequently, the additional constraints are removed to broaden the scope of the search, which yields 15 papers, that meet the criteria. Research gaps and limitations are instead identified through the analysis of the expanded set of publications retrieved using the less restrictive search string. RQ2 is also not restricted, as it represents an independent question requiring more detailed analysis. After these considerations, the final queries are chosen as such:

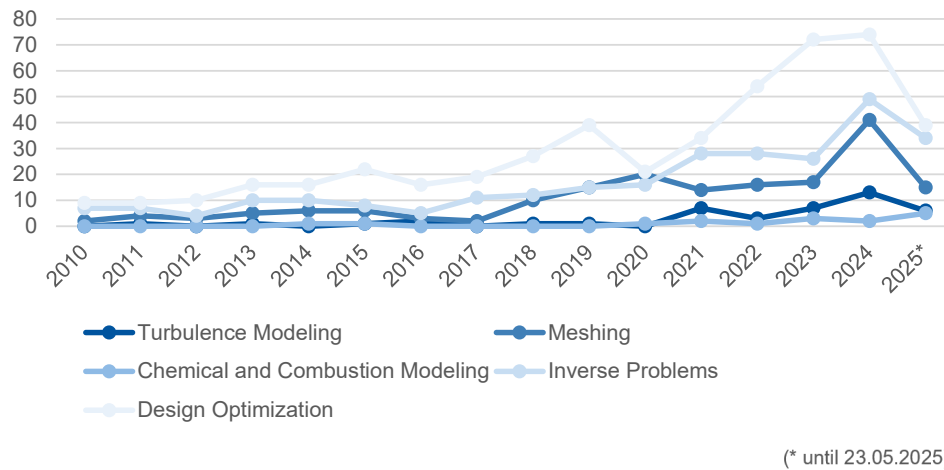


Figure 3.5: Number of found references (with distinction of research area) over the years with the general search string

```
( TITLE-ABS-KEY ( "computational fluid dynamics" OR CFD OR "flow simulation*" OR
"liquid simulation*" OR "gas simulation*" OR "fluid simulation*" OR "numerical fluid
mechanics" OR "fluid flow computation" OR "computational flow analysis" OR "simulation
of fluid dynamics" OR "flow model*" OR "fluid dynamics simulation*" ) )

AND ( TITLE-ABS-KEY ( "machine learning" OR "artificial intelligence" OR "deep
learning" OR "deep neural network*" OR "reinforcement learning" OR "deep reinforcement
learning" OR "physics informed neural network*" OR "physics informed machine learning"
OR "Bayesian optimization" OR "transfer learning" OR "generative AI" OR "convolutional
neural network*" OR "recurrent neural network*" OR pinn* OR metamodel* OR "data-based
model*" OR "neural network*" OR "surrogate model*" OR "pattern recogn*" OR "predictive
model*" OR "parameter tun*" OR "reduced order model*" ) )

AND ( TITLE-ABS-KEY ( "turbulence model*" OR "turbulence closure model*" OR "turbulence
closure" OR "Reynolds-averaged Navier-Stokes" OR "RANS" OR "Large Eddy Simulation*" OR
"LES" OR "Direct Numerical Simulation*" OR "DNS" OR "eddy viscosity model*" OR "second-
moment closure" OR "Reynolds stress model*" OR "RSM" OR "hybrid RANS-LES" OR "Detached
Eddy Simulation*" OR "DES" OR "scale-adaptive simulation*" OR "SAS" OR "k-epsilon
model*" OR "k-omega model*" OR "Spalart-Allmaras" OR "transition model*" OR "flow
separation model*" OR "turbulent flow simulation*" ) )

AND ( TITLE-ABS-KEY ( "gap*" OR "deficien*" OR "absen*" OR "unaddressed" OR "missing"
OR "lack" OR "insufficien*" OR "uninvestigated" OR "rarely addressed" OR "limitation*"
OR "underexplor*" OR "unresolv*" OR "research gap*" OR "future research*" OR "need for
research" OR "future work*" OR "further investigat*" OR "subsequent research*" OR
"future stud*" OR "research necessit*" OR "need for study" OR "ongoing research*" OR
"next step*" OR "follow-up" OR "extension of current work*" ) )

AND ( TITLE-ABS-KEY ( challenge* OR barrier* OR obstacle* OR limitation* OR difficult*
OR hurdle OR complication* OR issue* OR problem* OR setback* OR drawback* OR
constraint* OR restriction* OR shortcoming* OR weakness* ) )

AND ( PUBYEAR > 2019 ) AND ( LIMIT-TO ( SRCTYPE , "j" ) ) AND ( LIMIT-TO ( SUBJAREA ,
"ENGI" ) OR LIMIT-TO ( SUBJAREA , "COMP" ) OR LIMIT-TO ( SUBJAREA , "MATH" ) ) AND (
LIMIT-TO ( LANGUAGE , "English" ) )
```

Figure 3.6: The turbulence modeling query

```
( TITLE-ABS-KEY ("computational fluid dynamics" OR CFD OR "flow simulation*" OR "liquid
simulation*" OR "gas simulation*" OR "fluid simulation*" OR "numerical fluid mechanics"
OR "fluid flow computation" OR "computational flow analysis" OR "simulation of fluid
dynamics" OR "flow model*" OR "fluid dynamics simulation*") )

AND ( TITLE-ABS-KEY ("machine learning" OR "artificial intelligence" OR "deep learning"
OR "deep neural network*" OR "reinforcement learning" OR "deep reinforcement learning"
OR "physics informed neural network*" OR "physics informed machine learning" OR
"Bayesian optimization" OR "transfer learning" OR "generative AI" OR "convolutional
neural network*" OR "recurrent neural network*" OR "PINN*" OR "metamodel*" OR "data-
based model*" OR "neural network*" OR "surrogate model*" OR "pattern recogn*" OR
"predictive model*" OR "parameter tun*" OR "reduced order model*") )

AND ( TITLE-ABS-KEY ("mesh refinement*" OR "mesh smoothing" OR "mesh adaptation" OR
"mesh construct*" OR "mesh generat*" OR "mesh morph*" OR "grid refinement*" OR "grid
smoothing" OR "grid adaptation" OR "grid construct*" OR "grid generat*" OR "grid
morph*" ) )

AND ( LIMIT-TO ( LANGUAGE, "English" ) ) AND ( PUBYEAR > 2017 )

AND ( LIMIT-TO ( SUBJAREA, "ENGI" ) OR LIMIT-TO ( SUBJAREA, "COMP" ) OR LIMIT-TO (
SUBJAREA, "MATH" ) )
```

Figure 3.7: The meshing query

```
( TITLE-ABS-KEY ( "computational fluid dynamics" OR CFD OR "flow simulation*" OR
"liquid simulation*" OR "gas simulation*" OR "fluid simulation*" OR "numerical fluid
mechanics" OR "fluid flow computation" OR "computational flow analysis" OR "simulation
of fluid dynamics" OR "flow model*" OR "fluid dynamics simulation*" ) )

AND ( TITLE-ABS-KEY ( "machine learning" OR "artificial intelligence" OR "deep
learning" OR "deep neural network*" OR "reinforcement learning" OR "deep reinforcement
learning" OR "physics informed neural network*" OR "physics informed machine learning"
OR "Bayesian optimization" OR "transfer learning" OR "generative AI" OR "convolutional
neural network*" OR "recurrent neural network*" OR pinn* OR metamodel* OR "data-based
model*" OR "neural network*" OR "surrogate model*" OR "pattern recogn*" OR "predictive
model*" OR "parameter tun*" OR "reduced order model*" ) )

AND ( TITLE-ABS-KEY ("chemical reaction model*" OR "combustion model*" OR "chemical
reaction*" OR "combustion*") )

AND ( TITLE-ABS-KEY ( "gap*" OR "deficien*" OR "absen*" OR "unaddressed" OR "missing"
OR "lack" OR "insufficien*" OR "uninvestigated" OR "rarely addressed" OR "limitation*"
OR "underexplor*" OR "unresolv*" OR "research gap*" OR "future research*" OR "need for
research" OR "future work*" OR "further investigat*" OR "subsequent research*" OR
"future stud*" OR "research necessit*" OR "need for study" OR "ongoing research*" OR
"next step*" OR "follow-up" OR "extension of current work*" ) )

AND ( TITLE-ABS-KEY ( challenge* OR barrier* OR obstacle* OR limitation* OR difficult*
OR hurdle OR complication* OR issue* OR problem* OR setback* OR drawback* OR
constraint* OR restriction* OR shortcoming* OR weakness* ) )

AND ( PUBYEAR > 2018 ) AND ( LIMIT-TO ( SRCTYPE , "j" ) ) AND ( LIMIT-TO ( SUBJAREA ,
"ENGI" ) OR LIMIT-TO ( SUBJAREA , "COMP" ) OR LIMIT-TO ( SUBJAREA , "MATH" ) ) AND (
LIMIT-TO ( LANGUAGE , "English" ) )
```

Figure 3.8: The chemical and combustion modeling query

```
( TITLE-ABS-KEY ("computational fluid dynamics" OR CFD OR "flow simulation*" OR "liquid
simulation*" OR "gas simulation*" OR "fluid simulation*" OR "numerical fluid mechanics"
OR "fluid flow computation" OR "computational flow analysis" OR "simulation of fluid
dynamics" OR "flow model*" OR "fluid dynamics simulation*" ) )

AND ( TITLE-ABS-KEY ("machine learning" OR "artificial intelligence" OR "deep learning"
OR "deep neural network*" OR "reinforcement learning" OR "deep reinforcement learning"
OR "physics informed neural network*" OR "physics informed machine learning" OR
"Bayesian optimization" OR "transfer learning" OR "generative AI" OR "convolutional
neural network*" OR "recurrent neural network*" OR PINN* OR metamodel* OR "data-based
model*" OR "neural network*" OR "surrogate model*" OR "pattern recogn*" OR pattern* OR
"predictive model*" OR "parameter tun*" OR "reduced order model*") )

AND ( TITLE-ABS-KEY ("inverse problem*" OR "inverse model*" OR "parameter identif*" OR
"parameter estimat*" OR "model calibrat*" OR "model fitt*" OR "model adapt*" OR "model
validat*" OR "adjoint*" OR "sensitivity analys*") )

AND ( TITLE-ABS-KEY ("gap*" OR "deficien*" OR "absen*" OR "unaddressed" OR "missing" OR
"lack" OR "insufficient" OR "stud*" OR "uninvestigated" OR "rarely addressed" OR
"limitation*" OR "underexplor*" OR "unresolv*" OR "research gap*" OR "future research*"
OR "need for research" OR "future work*" OR "further investigat*" OR "subsequent
research*" OR "future stud*" OR "research necessit*" OR "need for study" OR "ongoing
research*" OR "next step*" OR "follow-up" OR "extension of current work*") )

AND ( TITLE-ABS-KEY (challenge* OR barrier* OR obstacle* OR limitation* OR difficult*
OR hurdle OR complication* OR issue* OR problem* OR setback* OR drawback* OR
constraint* OR restriction* OR shortcoming* OR weakness* OR gap* OR deficien* OR absen*
OR unaddressed OR missing OR lack OR uninvestigated OR "rarely addressed" OR
underexplor* OR unresolv* OR "research gap*" OR "future research*" OR "need for
research*" OR "future work" OR "further investigat*" OR "subsequent research*" OR
"future stud*" OR "research necessity" OR "need for study" OR "ongoing research" OR
"next step*" OR follow-up OR "extension of current work" ) )

AND ( PUBYEAR > 2017 ) AND ( LIMIT-TO ( LANGUAGE, "English" ) ) AND ( LIMIT-TO (
SUBJAREA, "ENGI" ) OR LIMIT-TO ( SUBJAREA, "COMP" ) OR LIMIT-TO ( SUBJAREA, "MATH" ) )
```

Figure 3.9: The inverse problems query

```
( TITLE-ABS-KEY ( "computational fluid dynamics" OR CFD OR "flow simulation*" OR
"liquid simulation*" OR "gas simulation*" OR "fluid simulation*" OR "numerical fluid
mechanics" OR "fluid flow computation" OR "computational flow analysis" OR "simulation
of fluid dynamics" OR "flow model*" OR "fluid dynamics simulation*" ) )

AND ( TITLE-ABS-KEY ( "machine learning" OR "artificial intelligence" OR "deep
learning" OR "deep neural network*" OR "reinforcement learning" OR "deep reinforcement
learning" OR "physics informed neural network*" OR "physics informed machine learning"
OR "Bayesian optimization" OR "transfer learning" OR "generative AI" OR "convolutional
neural network*" OR "recurrent neural network*" OR pinn* OR metamodel* OR "data-based
model*" OR "neural network*" OR "surrogate model*" OR "pattern recogn*" OR "predictive
model*" OR "parameter tun*" OR "reduced order model*" ) )

AND ( TITLE-ABS-KEY( "design optimization" OR "design process*" OR "shape optimization"
OR "geometry optimization" OR "topology optimization" OR "robust design*" OR "design
exploration" ) )

AND ( PUBYEAR > 2022 ) AND ( LIMIT-TO ( SRCTYPE , "j" ) ) AND ( LIMIT-TO ( SUBJAREA ,
"ENGI" ) OR LIMIT-TO ( SUBJAREA , "COMP" ) OR LIMIT-TO ( SUBJAREA , "MATH" ) ) AND (
LIMIT-TO ( LANGUAGE , "English" ) )
```

Figure 3.10: The design optimization query

3.2.2 Screening

Screening involves reviewing the titles and abstracts of all remaining records to assess their relevance to the research questions. Additionally, any papers that are deemed potentially relevant but whose full texts are not accessible are recorded separately, and efforts are made to retrieve them through academic networks or institutional access. The number of studies excluded at this stage is documented, along with the number that cannot be retrieved.

3.2.3 Eligibility

In the Eligibility phase, the remaining studies undergo a detailed full-text review. At this stage, predefined eligibility criteria are applied to determine whether a study should proceed to full-text evaluation. In this review, these criteria include the presence of AI integration into CFD, relevance to one of the defined thematic areas (e.g., turbulence modeling, mesh generation), relevance in the engineering domain, and the proposal of a novel methodology.

3.2.4 Inclusion

At the end of the PRISMA flow diagram, the Inclusion phase lists the number of studies that satisfy all criteria and are ultimately included in the systematic review. These studies form the basis of the subsequent quantitative and qualitative evaluation.

3.3 Quantitative Analysis

The quantitative analysis focuses on measurable parameters derived directly from the reported results in the reviewed literature. These parameters enable an objective comparison of methods based on their empirical performance, scope, and evaluation rigor.

Table 3.2: Parameters

AI/ML Model	Identifies the specific artificial intelligence or machine learning approach used (e.g., convolutional neural network, random forest, physics-informed neural network), allowing trends in methodological adoption to be tracked.
CFD Problem Type	Specifies the category of CFD flow (e.g. steady, 2D, incompressible, subsonic), providing context for performance comparisons.
Validation Method	Describes how the model's performance was evaluated (e.g., comparison to experimental data, high-fidelity simulations, benchmark datasets), giving insight into the robustness of reported results.

3.4 Qualitative Analysis

The qualitative criteria capture conceptual, interpretative, and workflow-related aspects of the reviewed references. The criteria are defined on a four-point scale ranging from “Very Low (1)” to “Very High (4)”.

General Criteria

The general criteria that are applied to all papers, as well as the definition of the four-point rating scales are as follows:

Table 3.3: General qualitative criteria

Criteria	Rating Scale
Novelty of methods The degree of innovation in the proposed AI approach	(1) Standard ML methods without any new ideas or adaptations (2) Minor adaption of existing methods (3) Proposes new architecture or combination or major adaptation (4) Proposes a novel paradigm or modeling philosophy that can be used as the new standard, conceptually original
Generalizability The applicability of the method is across different flow types, geometries, or boundary conditions	(1) Highly specific and not (yet) generalizable (2) Limited applicability for few use cases (3) Applicable to diverse cases with limitations (4) Generalizable to almost all use cases
Computational Efficiency Efficiency compared to traditional CFD solver	(1) Slower than traditional methods (2) Similar speed compared to traditional methods (3) Slightly faster than traditional methods (4) Significantly faster than traditional methods
Validation The extent and credibility of the model's evaluation	(1) No known validation or lacking validation (2) Validated on limited academic benchmarks (3) Validated on multiple standard flows (4) Extensive validation across reproducible industrial and academic datasets
Integration and Automation in CFD-Workflow Evaluation of how well the AI method integrates into the existing CFD pipeline and to what extent it automates parts of the process	(1) Almost no automation (2) Partial AI support, but manual steps still dominate, no adaptive behavior, such that human intervention can be needed in change of conditions (3) Model well integrated into solver frameworks, supports adaptivity (4) Full pipeline automated

Domain-specific Criteria

In addition to the general evaluation criteria, domain-specific criteria are chosen to better represent the different areas of research.

Table 3.4: Turbulence modeling qualitative criteria

Robustness Stability under numerics, evaluation of how reliably a turbulence model performs across different meshes, and timestep conditions	(1) Model prone to numerical instability or divergence in many flow cases (2) Model sensitive to mesh, boundary conditions or initial guesses (3) Model mostly stable, with occasional need for tuning or grid refinement (4) Model highly stable in diverse flow types, even with coarse grids
Physical Basis Evaluation of how well a turbulence model is grounded in known physical laws and fluid dynamics principles, the model's interpretability and trustworthiness in academic and industrial contexts.	(1) Weak physical assumptions, mostly fitted to specific experiments (2) Heavily empirical, limited theoretical justification, but works well in practice (3) Based on physical reasoning with some empirical adjustments (4) Strong foundation in turbulence physics, derived from rigorous modeling assumptions

Table 3.5: Meshing qualitative criteria

Mesh Quality The quality of the generated or optimized mesh in comparison to traditional meshing methods	(1) Worse than traditional meshing methods (2) Similar quality to traditional methods (3) Slightly improved quality over non-AI methods (4) Highly improved quality compared to traditional methods
Mesh Improvement over Baseline Level of enhancement over the baseline mesh, applies specifically to methods that take an initial mesh as an input and aim to improve it.	(1) Little to no measurable improvement over the initial mesh. (2) Noticeable improvements in certain regions or metrics, but with limited global impact or no impact on simulation results. (3) Clear and measurable improvement across multiple mesh quality metrics or a demonstrated impact on simulation performance (4) Strong, physics-aware improvement with high impact on simulation performance

Table 3.6: Chemical and combustion modeling qualitative criteria

Chemical fidelity in AI-assisted modeling Evaluation of how well an AI-assisted model reproduces the underlying chemical and combustion processes in reactive flows	(1) Ignores chemical kinetics or reduces to overly simplified models (2) Limited inclusion of reaction mechanisms, e.g. only global models (3) Incorporates multi-step kinetics or surrogate reaction schemes (4) Accurately models detailed chemistry or uses AI to accelerate stiff detailed mechanisms while preserving fidelity
---	--

Table 3.7: Inverse problems qualitative criteria

Uncertainty Handling The ability to represent and manage uncertainty during the reconstruction of unknown inputs and to distinguish between different sources of uncertainty	(1) No treatment of uncertainty or multiple solutions (2) Basic heuristics for uncertainty or regularization (3) Partial UQ or probabilistic inference used (4) Robust UQ with well-characterized uncertainty and stability
--	--

Table 3.8: Design optimization qualitative criteria

Design Quality The practicality, realism, and engineering value of designs generated through AI-assisted optimization	(1) Designs are unrealistic, non-manufacturable, or physically invalid, little to no validation of design feasibility or performance (2) Designs are plausible but only validated on idealized or simple configurations, questionable practical use (3) Physically meaningful and potentially manufacturable designs, validation is on relevant test cases, performance gains relative to baseline configurations or conventional optimization methods (4) Designs with high practical quality: innovative, realistic, and directly applicable in engineering practice, constraints (aerodynamic, structural, thermal, manufacturing, etc.) explicitly incorporated, competitive or superior results compared to classical methods.
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4 Results

4.1 Results of the Inclusion Process

In Figure 4.1, the results of the inclusion processes for the five research fields are shown. The stepwise narrowing illustrates how the systematic review process filters a large initial set of records down to a smaller, high-quality selection suitable for analysis.

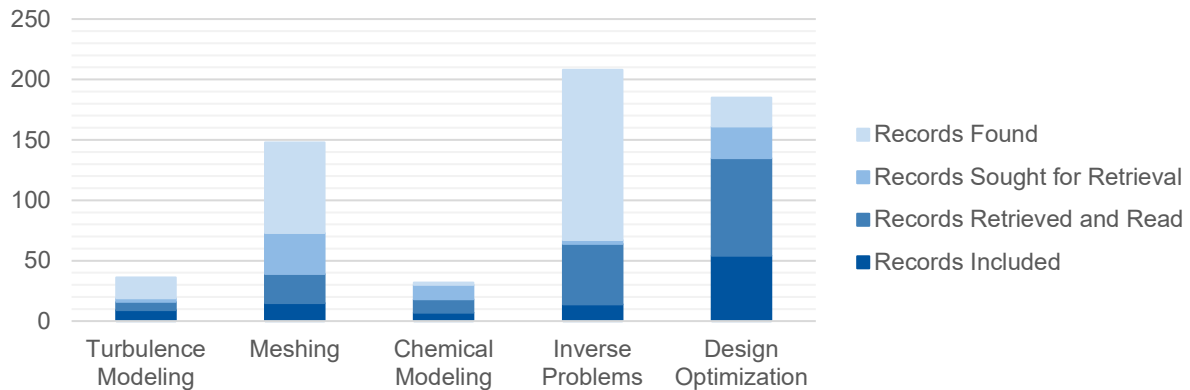


Figure 4.1: Inclusion process of the records with the PRISMA Framework

Table 4.1: Number of records in every step of the inclusion process

Research Area	Found	Sought for Re- trieval	Retrieved and Read	Included
Turbulence Modeling	36	19	16	9
Meshing	148	73	39	15
Chemical and Combustion Modeling	21	12	11	7
Inverse Problems	208	67	64	14
Design Optimization	185	161	135	54

4.2 Quantitative Analysis

4.2.1 Turbulence Modeling

As demonstrated in Chapter 2, conventional turbulence closure methods pose many challenges and limitations. The methods included after the PRISMA selection process (Table 4.2) all make use of different ML approaches, and even paradigms (supervised learning vs reinforcement learning), they are explained separately, with focus on their distinctive features.

The earliest included work is from ZHU ET AL. (2019) [Zhu19]. A distinctive contribution by this paper lies in the pioneering idea of departing from the conventional paradigm of augmenting or correcting RANS models, as done by PARISH & DURASAMY [Par16].

Table 4.2: Overview turbulence modeling papers

Reference	AI/ML Algorithm	CFD Problem Type
[Zhu19]	Radial Basis Function Neural Network (RBFNN)	RANS compressible 2D subsonic, high Re
[Nov20]	Multi-Agent Reinforcement Learning (MARL)	LES incompressible 3D forced isotropic turbulence
[Sto21]	Artificial Neural Network (ANN)	LES incompressible 3D isothermal
[Lee23]	Artificial Neural Network-based Wall Stress Model (AWM)	LES incompressible 3D
[Zha24c]	Convolutional Neural Network (CNN) (3D U-Net)	RANS incompressible trained on 2D, validated on 3D
[Fox24]	Multiple variants of Physics-Informed Neural Networks (PINNs)	RANS incompressible 2D
[Di24]	Gene Expression Programming (GEP)	RANS most incompressible, one case compressible 2D
[Fan25]	ConvPDE, ConvLSTM and Diffusion-based Generative Model (DGM)	DNS incompressible 2D

Rather than introducing corrections to existing PDE-based closures, a fully data-driven approach is developed, which reconstructs an algebraic mapping between the turbulent eddy viscosity μ_t and mean flow variables using a NN architecture. This modeling strategy is applied to high Reynolds number flows, which is a regime characterized by strong anisotropy and large-scale variations. As mentioned in Chapter 2, high Reynolds number flows pose significant challenges for generalization and robustness. The authors divide the flow domain into four physically motivated regions: the near-wall layer, wake region, far-field, and transitional zone. Then, separate radial basis function neural networks (RBFNNs) for each zone are trained. RBFNNs are artificial networks with a radial basis function as the activation function, the value of which only depends on the distance from a fixed point [Moh16].

In another pioneering study, NOVATI ET AL. (2020) [Nov20] introduce a reinforcement learning (RL) framework for SGS turbulence modeling in LES, thereby addressing critical limitations of supervised learning methods. Supervised learning-based closures often suffer from poor generalization beyond the training data and are constrained by the requirement for differentiable targets derived from high-fidelity DNS data. To overcome these challenges, a novel formulation as a multi-agent reinforcement learning (MARL) problem is proposed, where multiple agents interact with the flow field and learn a control policy to locally adapt the dimensionless coefficient C_s from the Smagorinsky SGS model. Remember and Forget Experience Replay (ReF-ER) ensures training stability and sample efficiency by making use of Experience Replay (ER), in which experiences are reused over multiple policy iterations.

STOFFER ET AL. (2021) [Sto21] proposes a further novel SGS model for LES, this time based on artificial neural networks (ANNs). Each ANN uses as input a local $5 \times 5 \times 5$ stencil of resolved velocity components u, v, w . Training data are generated from DNS of turbulent channel flow at $Re = 590$ that are filtered using a discrete top-hat filter to mimic coarse LES fields. By comparing the resulting low-resolution fluxes with the true DNS fluxes, the authors extract SGS correction terms that reflect both unresolved turbulence and numerical discretization errors.

Wall modeling is an essential component of LES to alleviate the need for extremely fine near-wall grids, which otherwise dominate the computational cost, particularly at high Reynolds numbers [Bos18]. However, existing wall-stress modeling approaches present significant limitations [Lee23]. As a data-driven approach, LEE ET AL. (2023) [Lee23] propose ANN-based Wall-Stress Models (AWMs) for Wall-Modeled Large-Eddy Simulations (WMLES) of turbulent channel and separated turbulent boundary layer (TBL) flows at high Reynolds numbers. AWMs avoid near-wall grid refinement and achieve Reynolds-number independent computational cost. The ANN has as output the skin-friction coefficient C_f , from which the wall-shear stress is computed as

$$\tau_w = \frac{1}{2} \rho U_b^2 C_f \quad (4.1)$$

with U_b as the bulk velocity and ρ the fluid density. The key novelty of this work is the use of inputs based on the Fukagata-Iwamoto-Kasagi (FIK) identity [Fuk02], which is an exact mathematical decomposition of the skin-friction coefficient C_f in wall-bounded flows into physically meaningful terms. These terms directly relate to known physical mechanisms in the flow, in contrast to the model having raw and potentially noisy near-wall data as input and learning abstract correlations.

ZHAO ET AL. (2024) [Zha24c] embed a convolutional neural network (CNN) into a RANS solver for urban wind simulations. The goal is to improve the accuracy of Reynolds stress prediction by incorporating spatial features through a CNN. This is achieved by applying convolutional filters (kernels) that slide across the input data, detecting local, spatial patterns and preserving their relative positions, such that patterns such as shear layers or eddies are recognized in the correct context. Also, since CNNs incorporate shared parameters, the same kernel is applied across the entire domain,

meaning the network can detect similar flow structures in different locations. Their framework, called CNN-RSM RANS, builds on POPE's tensor basis decomposition (1975) [Pop00], as discussed in Chapter 2.

FOX ET AL. (2024) [Fox24] conduct a systematic evaluation of several PINNs for turbulence modeling using the periodic hill benchmark. The PINN variants each reflect different levels of physical constraint enforcement within the RANS framework. These included the Direct Reynolds Stress Model (DRSM), which predicts first-order Reynolds stresses to close the system; Reduced Boundary Models (RBMuvp and RBMuv), which omit pressure and/or stress terms from boundary enforcement; and the Continuity-Only Model (COM), which solves the flow using only the continuity equation. COM is a novel formulation introduced and evaluated by the authors. Three additional models incorporated turbulence closure assumptions: the Mixing Length Model (MLM), using LUDWIG PRANDTL's Mixing Length Model approach [Pra25] to estimate turbulent viscosity; the Turbulent Viscosity Model (TVM), which directly predicts the turbulent viscosity μ_t and the Turbulent Viscosity and Kinetic Energy Model (TVKEM), which, in addition to the turbulent viscosity, also predicts k to complete the Boussinesq approximation [Bou77] without simplifications.

To improve one-equation RANS turbulence models, DI FABBIO ET AL. (2024) [Di24] apply Gene Expression Programming (GEP) to the Spalart–Allmaras (SA) SGS model. GEP is an evolutionary algorithm that evolves model terms using genetic operations like mutation and crossover [Fer01]. The study of DI FABBIO ET. AL. adds extra anisotropy terms to the Reynolds stress tensor and models anisotropic stresses as weighted combinations of Pope-based tensor bases, as discussed in Chapter 2, and scalar invariants. The authors also derive the Coupled Spalart–Allmaras (CSA) model, combining elements from SA and k – ω models. CSA features modified destruction and production terms of the SA model; the former of which remains active away from walls, and the latter in regions of high strain or vorticity. They propose and test the following combinations: SA with extra anisotropy terms, SA with modified destruction/production terms, CSA (that already implements modified destruction/production terms), and CSA with extra anisotropy terms.

Lastly, FAN ET AL. (2025) [Fan25] present a novel two-stage turbulence modeling framework. In the first stage of their approach, a hybrid differentiable neural solver operates on a coarse grid to resolve large-scale turbulent structures. Here, two architectural designs are explored: NeuralPDE-Corr, which embeds ConvLSTM-based correction terms based on CNNs and LSTMs, and NeuralPDE-Discretize, which parameterizes the interpolation and discretization schemes as trainable NNs. The second stage employs a Bayesian diffusion model that generates small-scale turbulence structures conditioned on the large-scale dynamics predicted by the coarse-grid solver. A key innovation of the framework lies in its temporal modeling capability by integrating ConvLSTM layers, which distinguishes it from static surrogate models that lack memory of past states.

Validation Results

The validation methods used in these references reveal several key trends. A priori validation, where models are evaluated offline using pre-existing datasets, is commonly performed using high-fidelity DNS data as a baseline, though some studies also utilize RANS baselines with various turbulence models [Zhu19; Zha24c] or experimental measurements [Di24]. On the other hand, a posteriori validation is conducted on LES simulation workflows [Sto21; Lee23]. In contrast to priori validation, a posteriori validation involves online testing, where the model is embedded directly into a CFD simulation to assess its performance in practice. This evaluates not only prediction accuracy but also the model's impact on solution stability, convergence, and physical realism within the full CFD pipeline. A posteriori testing is therefore more indicative of real-world applicability.

Furthermore, a consistent emphasis is placed on testing generalization to high Reynolds number flows, which stems from the reasons why current turbulence closure models lack generalizability, as explained more in detail in Chapter 2. Furthermore, most works employ distinct datasets for training and testing to validate the generalizability. The selected test cases often involve geometries that induce turbulence and flow separation, such as periodic hills [Fox24] and backward-facing steps [Zha24c], the latter of which is characterized by adverse pressure gradients. Some references also include realistic industrial configurations that support comprehensive a posteriori assessment [Zha24c; Zhu19; Di24]. Especially ZHAO ET. AL. [Zha24c] include comprehensive validation by testing on an isolated building, a building array, and the building complexes in Niigata, Japan.

4.2.2 Meshing

Mesh generation

Traditional mesh generation algorithms face several limitations: they are often slow, require substantial manual intervention, and typically lack the ability to incorporate flow-specific features (e.g. shock waves) that demand locally refined meshing. AI-based methods are used to overcome these problems and to improve the meshing process in the CFD pipeline. Table 4.3 provides an overview of the studies reviewed in this work, summarizing the algorithms employed and the types of flow scenarios in which the meshing approaches are applied.

In the following the above-mentioned papers and their methods are presented in more detail:

An improved version of the advancing front technique (AFT) is developed to address convergence problems while reducing computation time by 40–60% through the integration of an ANN trained on high-quality solutions. Although the resulting meshes achieve substantial time savings, they still require additional optimization and smoothing to reach the desired quality [Liu24c]. These phenomena, such as shock waves,

Table 4.3: Overview mesh generation papers

Reference	AI/ML algorithm	CFD problem type
[Lim24]	DRL	boundary layer of an airfoil (2D)
[Liu24c]	ANN	aircraft shape (2D) and simple shapes in 3D
[Loc23]	ANN	airfoil (2D)
[LU22]	ANN	airfoil and aircraft model (2D)
[Pan23a]	RL	Complex shapes (2D)
[Pel24]	GNN	Deforming plate, flow around cylinders (2D)
[San24]	ANN	Airfoil (2D)

require a higher mesh resolution. This predictive capability enables targeted refinement, though the produced meshes may also need further improvement [Loc23]. Building on the latter strategy, another study tackles the challenge of stretched elements by using an ANN trained on high-fidelity CFD simulations to generate an initial mesh. This mesh can serve directly as the starting point for adaptive mesh refinement (AMR), effectively decreasing the number of AMR iterations [San24]. In the context of anisotropic grids for airfoil boundary layers, a dual-ANN setup is proposed. One network predicts the advancing direction for grid construction starting from the airfoil and the other estimates the appropriate element size [LU22].

A different approach, proposed by PELISSIER ET AL., applies two GNNs in sequence. The first learns mesh parameters typical of open-source mesh generators to create an initial mesh, while the second adapts this mesh using information from simulation results. This two-stage process delivers a speedup of 63.4 times compared to traditional meshing methods [Pel24].

LIM ET AL. develop a novel approach by employing RL to the boundary layer meshing problem. Here, an agent incrementally generates layers starting from the airfoil surface, guided by a reward function based on mesh quality metrics [Lim24].

Mesh Smoothing

Mesh smoothing refers to the process of repositioning mesh nodes to enhance overall mesh quality to improve the result of the CFD simulation. This quality is typically quantified using several metrics, including element aspect ratio, which favors geometrically regular elements, and skewness, which measures deviation from ideal element shapes [Wan24a]. A third important metric is orthogonality, which is a measure for how close the cell edges are to being orthogonal to the vector from the cell-center to the edge

midpoint [Wan25a]. An overview of the approaches included in this study is provided in Table 4.4.

Table 4.4: Overview mesh smoothing papers

Reference	AI/ML algorithm	CFD problem type
[Wan25a]	GNN	Airfoil, channel flow (2D)
[Wan24a]	DRL	Cylinder, airfoil (2D)

WANG ET AL. represent meshes as graphs and apply a GNN to perform smoothing without relying on high-quality training data, as the loss function is derived directly from a mesh quality measure. Although the method does not consistently exceed the quality of optimization-based smoothing, it achieves a speedup of 13.56 times [Wan25a].

A separate approach uses DRL with a reward function based on mesh quality metrics to guide the smoothing process. This method is 38.88 times slower than Laplacian smoothing but remains 11.09 times faster than optimization-based techniques. Its main advantage is producing, on average, higher quality meshes than heuristic, optimization-based, and neural network-based smoothing methods [Wan24a].

Mesh adaptation

Mesh adaptation is applied to an initial mesh to better resolve flow features that necessitate localized refinement. Traditionally, an iterative and computationally intensive process, mesh adaptation can be accelerated through the use of AI-based techniques. Two primary adaptation strategies are commonly employed: H-type adaptation, which involves locally adding or removing elements, and R-type adaptation, which adjusts the positions of existing nodes while keeping the total number of elements constant [Mia23]. Table 4.5 summarizes the proposed mesh adaptation methods reviewed in this study.

Table 4.5: Overview mesh adaptation papers

Reference	AI/ML algorithm	CFD problem type	
[Fou23]	DRL	Advection (1D and 2D)	H-type
[Mia23]	U-NET	Nuclear reactor channel (2D)	
[Obi23]	DL	Cylinder, airfoil (2D)	
[Pei24]	GNN	Deforming plate, flow around cylinders (2D)	R-type
[WU22]	ANN	Cylinder, airfoil (2D)	
[Yu25]	Diffusion	Airfoil, missile (2D)	
[Yu23]	MLP, SOM	Airfoil (2D)	

MIAO ET AL. employ a U-NET to identify regions requiring refinement directly from an image of the initial coarse mesh, avoiding both CFD simulations and explicit error

estimation. U-NET is a neural network architecture specifically designed to have images both as input and output. Once refinement areas are determined, additional nodes are inserted and smoothed using conventional techniques [Mia23].

OBIOLS ET AL. develop a DNN whose loss function incorporates both the data error on the mesh and the PDE residual, thereby including information about the physics of the underlying problem without requiring domain-specific knowledge. From a low-resolution input mesh and its initial solution, the DNN produces a refined mesh that matches the quality of traditional iterative AMR methods while being 3.2 to 5.5 times faster and eliminating the need for iterative refinement [Obi23].

A different strategy applies DRL to refine or coarsen the mesh locally, guided by a reward function that balances solution accuracy against the computational cost of running the CFD simulation. This method requires no training data and yields meshes that are coarser than those from traditional AMR, yet with smaller solution errors and lower computational cost of obtaining the solution [Fou23].

WU ET AL. increase mesh density in regions of high gradients identified from an initial CFD simulation and then update the mesh iteratively using an ANN, that predicts the flow solution on each updated mesh. This approach avoids running a full CFD simulation at each update step and performs a final simulation only once the mesh aligns with the flow characteristics [WU22].

YU ET AL. propose a method that first runs a CFD simulation on an initial mesh and then applies a trained MLP to detect regions of rapid change where a higher resolution mesh is needed. Unlike traditional approaches relying solely on Hessian or gradient information, this method maintains accuracy on low-quality input meshes. A self-organizing mechanism (SOM) moves nodes under smoothing constraints to preserve mesh quality. An SOM is an unsupervised neural network that projects high-dimensional data onto a lower-dimensional grid while preserving topological relationships between data points [Yu23]. In a later work, YU ET AL. introduce a new architecture based on two diffusion models. The first extracts flow features from design parameters, while the second moves nodes according to the predicted features. Diffusion models work by first adding noise to the training data and then learning to reverse this to recover the data, allowing the trained models to generate data from random noise. This approach achieves roughly twice the speed of the SOM-based method while maintaining comparable mesh quality [Yu25].

4.2.3 Chemical and Combustion Modeling

AI-based methods for chemical and combustion modeling in CFD aim to address some of the most persistent challenges in the field: the high computational cost of detailed chemical kinetics, the complexity of coupling multiple physical processes such as turbulence, radiation, and multi-species transport, and the scarcity of experimental validation data. The studies reviewed here employ a wide range of machine learning architectures, spanning from classical neural networks to hybrid physics-informed frameworks, applied to diverse combustion regimes and operating conditions.

The first category of approaches focuses on reduced-order modeling of combustion dynamics, particularly in systems characterized by strong unsteadiness. MENDIBLE ET AL. [Men21] employ a suite of data-driven reduced-order modeling techniques to analyze unsteady combustion dynamics in rotating detonation engines (RDEs). These methods include UnTWIST (Unsteady Time-Series Identification of Sparse Transitions), Sparse Relaxed Regularized Regression (SR3), Lotka-Volterra model, Dynamic Mode Decomposition (DMD), and Koopman operator forecasting.

UnTWIST is a sparse regression-based algorithm designed to discover governing equations directly from time-series data. Its goal is to identify a sparse coefficient vector ξ such that:

$$\dot{X} \approx \theta(X)\xi, \quad (4.2)$$

where $\dot{X}$ is time derivative of X , and $\theta(X)$ is library of candidate functions.

Similarly, SR3 enforces sparsity to identify dominant dynamical modes by solving an optimization problem with L1 regularization:

$$\min_{\beta} \{\|y - X\beta\|^2 + \lambda\|\beta\|^1\}, \quad (4.3)$$

where y is the observed data, X is the feature matrix, β are regression coefficients, λ controls the sparsity.

An additional nonlinear benchmark system that is often employed to evaluate data-driven combustion forecasting is the Lotka-Volterra system, also known as the predator-prey model. It describes the interaction between two coupled populations, prey $x(t)$ and predators $y(t)$ through:

$$\dot{x} = \alpha x - \beta xy, \quad (4.4)$$

$$\dot{y} = \delta xy - \gamma y, \quad (4.5)$$

where $x(t)$ is the prey population, $y(t)$ is the predator population, α is the prey growth rate, β is the predation rate coefficient, γ is the predator death rate, δ is the growth rate of predators per prey eaten. Though originally ecological, this system serves as a convenient reduced-order test case for nonlinear dynamics relevant to combustion oscillations.

Other methods rely on spectral decompositions of dynamical systems. DMD extracts coherent spatio-temporal structures from high-dimensional flow data by approximating the linear operator that maps snapshots of the system,

$$X' \approx A X \quad (4.6)$$

$$A = X' X^\dagger, \quad (4.7)$$

where $X^\dagger$ is the pseudoinverse of X .

Extending this concept, Koopman operator-based forecasting provides a way to represent nonlinear dynamics with a linear operator K acting in a higher-dimensional observable space,

$$K g(x) = g(f(x)), \quad (4.8)$$

where $f(x)$ is the nonlinear system dynamics, and g are observables mapping the state to the lifted space.

Building on these dynamical system approaches, other works emphasize probabilistic and optimization-based machine learning. For example, Gaussian Deep Neural Networks (GDNNs) [Xue23], which couple deep learning with probabilistic Gaussian layers to quantify predictive uncertainty, improving robustness across varying operating conditions, and Artificial Neural Networks coupled with Genetic Algorithms (ANN+GA) [Jai21], where GAs optimize network parameters for combustion efficiency and emissions prediction in internal combustion engines.

A third research direction leverages classical supervised learning architectures for combustion property prediction. ZHANG ET AL. [Zha25b] investigate multiple ML models for combustion property prediction, including Backpropagation Neural Networks (BPNs), Convolutional Neural Networks (CNNs), and Random Forests (RF). BPNs rely on gradient-based weight updates to minimize error, CNNs exploit convolutional filters to extract spatial features in flow and flame data, and RFs aggregate many decision trees to provide robust, ensemble-based predictions.

Complementing purely data-driven models, physics-informed architectures explicitly embed governing equations into the learning process. In the domain of radiative heat transfer, CHEN ET AL. [Che25a] propose a physics-informed neural network (PINN) architecture with dual MLPs. By embedding the radiative transfer equation into the loss function, PINNs ensure physical consistency while predicting emissivity and absorption coefficients, providing accuracy comparable to line-by-line (LBL) calculations at a fraction of the computational cost.

Further specialization emerges in domain-tailored surrogate models for combustion chemistry. CAI ET EL. [Cai25] introduce multi-layer perceptron (MLP) surrogates: ODE-MLP for solving ordinary differential equations of chemical kinetics and Therm-MLP for predicting thermochemical properties such as heat capacities and species enthalpies.

At the scale of industrial combustion systems, hybrid strategies that couple dimensionality reduction with ML regression are proven particularly effective. WANG ET EL. [Wan25c] combine Proper Orthogonal Decomposition (POD) with Extreme Gradient Boosting (XGBoost). POD reduces the high-dimensional flow field into dominant modes, which are then mapped to combustion outcomes using XGBoost regression. The approach is applied to coal-fired boiler combustion, with two modeling scales: Comprehensive Fuel Combustion Flow Dynamics (CFCFD), covering multiphysics interactions of flow, heat transfer, and combustion, and Full Combustion Flow Dynamics (FCFD), resolving spatial and temporal details of the process.

Across these studies, predictive accuracy is consistently evaluated using standard performance metrics. These include the Mean Absolute Error (MAE), representing average error magnitude, the Mean Absolute Percentage Error (MAPE), expressing error relative to true values, the Root Mean Square Error (RMSE), which penalizes large

deviations more heavily, the Coefficient of Determination (R^2), quantifying the variance explained by the model, and the Normalized RMSE (NRMSE), which rescales RMSE relative to data mean or range. In combustion engine studies, domain-specific measures such as Gross Indicated Efficiency (GIE) and Brake Mean Effective Pressure (BMEP) are also reported. GIE represents the ratio of in-cylinder indicated work to the fuel's chemical energy input, while BMEP reflects torque output normalized by displacement volume, serving as a standard benchmark in engine research [Jai21].

An overview of the methods and CFD problem types is provided in Table 4.6.

Table 4.6: Overview of the AI algorithms used and the specific CFD problem type

Reference	AI/ML algorithm	CFD problem type
[Men21]	UnTWIST, Sparse Relaxed Regularized Regression, Lotka-Volterra, DMD, Koopman operator forecasting	Rotating detonation engine flow field dynamics
[Jai21]	ANN + GA	3D RCCI combustion in a single-cylinder diesel engine
[Xue23]	Gaussian deep neural network (GDNN)	3D combustion temperature field prediction
[Zha25b]	BPNN, CNN, RF	Non-premixed combustion jet fire simulation of hydrogen blended natural gas pipelines
[Che25a]	PINNs with dual MLPs	Radiative heat transfer modeling in high-temperature reactive multi-species gas mixtures
[Cai25]	MLP surrogate models: Therm-MLP and ODE-MLP	Supercritical CH_4 premixed flames in isotropic turbulence and H_2/O_2 coaxial jet diffusion flames
[Wan25c]	POD + XGBoost	Coal-fired boiler combustion and flow field simulation (CFCFD and FCFD)

Validation

Validation across studies employs a wide range of experimental and numerical approaches, often combining high-fidelity CFD with physical measurements to ensure realism. MENDIBLE ET AL. [Men21] use high-speed imaging of rotating detonation engine combustion waves alongside CFD comparison and interpretability analysis. JAIN ET AL. [Jai21] validate their RCCI diesel engine predictions experimentally under two BMEP conditions, directly comparing ANN+GA results against CFD outputs. XUE ET AL. [Xue23] perform cross-validation over a range of burner angles, corroborated by CFD simulations and real plant thermocouple measurements. ZHANG ET AL. [Zha25b] benchmark their models using both experimental and CFD datasets for hydrogen-blended natural gas jet fires. CHEN ET AL. [Che25a] employ high-fidelity spectral calculations, analytical solutions, and FSCK model benchmarks to verify radiative transfer predictions. CAI ET AL. [Cai25] integrate direct numerical solutions of chemical kinetics with CFD simulations and experimental OH layer thickness measurements for premixed and diffusion flames. Finally, WAN ET AL. [Wan25c] validate their boiler optimization

framework using field test data in combination with CFD simulations, ensuring agreement across key process variables.

4.2.4 Inverse Problems

Unlike the forward problems in turbulence modeling, chemical or combustion modeling, where the governing equations and boundary conditions are assumed known, inverse formulations are typically ill-posed, requiring regularization or physics-guided learning methods to obtain stable solutions.

The performance of inverse methods is usually benchmarked against reference CFD or experimental data using relative error measures. For a predicted field u_{pred} and reference field u_{ref} , the relative error can be written as

$$Rel. Error = \frac{u_{pred} - u_{ref}}{u_{ref}} \quad (4.9)$$

while the Relative Root Mean Square Error (RRMSE), often used in reconstruction tasks, is defined as

$$RRMSE = \sqrt{\frac{\sum_{i=1}^N (u_i^{pred} - u_i^{ref})^2}{\sum_{i=1}^N (u_i^{ref})^2}} \quad (4.10)$$

The first example is surrogate-assisted calibration. CHOWDHARY ET AL. combined Proper Orthogonal Decomposition (POD) with machine learning regressors to build reduced-order surrogates of hypersonic flows, which were then embedded in a Bayesian inference framework. This enabled efficient calibration of SST turbulence model parameters against HIFiRE-1 experimental data. In Bayesian calibration, the posterior distribution of parameters θ given data y is expressed as

$$p(\theta | y) \propto p(y | \theta) p(\theta) \quad (4.11)$$

where $p(y|\theta)$ is the likelihood obtained from the surrogate model and $p(\theta)$ the prior. With this approach, parameter recovery errors were below 3%, while computational cost was reduced by several orders of magnitude compared to direct CFD [Cho22].

Physics-Informed Neural Networks (PINNs) provide a second route by embedding conservation laws into the training process. SEOHEE JANG and MOHAMMAD JADIDI applied a domain-decomposed PINN with Reynolds-averaged Navier-Stokes (RANS) closure to turbulent flow over porous media. Their training objective combined data-driven and physics-based losses. The approach achieved relative L_2 errors of 5.44% for velocity and 18.9% for turbulent kinetic energy, while hidden velocity magnitudes at unseen Reynolds numbers were recovered with differences of 8.55% and 4.39% [Jan24].

WANG introduced a physics-informed recurrent generative adversarial network (PIRGAN) for reconstructing high-resolution flow fields in rotating detonation combustors from sparse sensor data. The architecture of PIRGAN comprises three sub-networks; temporal differentiation network $TNet_{\theta_T}$, the reconstruction network $RNet_{\theta_R}$ and the discriminator network $DNet_{\theta_D}$. Their interaction follows a three-step recurrent

process. First, $TNet_{\theta_T}$ computes the temporal derivatives from the low-resolution input at the current timestep and the previous one. Second, these derivatives are combined with the last super-resolved output to generate a provisional high-resolution estimate. Third, $RNet_{\theta_R}$ refines this estimate into the final reconstruction, while $DNet_{\theta_D}$ evaluates whether the generated field is consistent with true high-fidelity CFD data. By enforcing physical constraints during adversarial training, the model achieved sub-1% RRMSE for detonation wave speed predictions while also capturing critical small-scale structures such as oblique shocks, slip lines, and contact surfaces that were previously invisible [Wan24b].

Paun L. developed the SECRET framework (Statistical Emulation for Computational Reverse Engineering and Translation), which integrates Gaussian process surrogates, deep residual neural networks, and Bayesian inference techniques. SECRET replaces the forward CFD solver with a statistical emulator, trained on a set of high-fidelity simulations, to approximate the system's response surface in parameter space. Once the emulator is constructed, inverse inference is carried out using a combination of complementary statistical techniques. Markov Chain Monte Carlo (MCMC) is employed to sample from the posterior distribution of the parameters, providing a probabilistic characterization of uncertainty. History Matching with rejection sampling serves as a filtering step, ruling out implausible regions of parameter space and thus reducing the computational burden of subsequent inference. Finally, simulated annealing is used as a global optimization strategy to efficiently explore complex, multimodal likelihood surfaces and avoid local minima. These complementary inference methods allow SECRET to explore the posterior distribution of uncertain physiological parameters and quantify their uncertainty with

$$p(\theta | y) \propto p(y | f_{\text{emu}}(\theta)) p(\theta) \quad (4.12)$$

where θ is the vector of unknown parameters and $f_{\text{emu}}(\theta)$ is the emulator prediction of the CFD model output for θ . The framework achieved low parameter errors in systemic circulation test cases with noise-free data down to $10^{-5}\%$ and maintained acceptable accuracy with 0.9-7% deviations with noisy inputs. However, posterior variance estimates deviated by 35-570% [Pau24].

MINGLIANG LI demonstrated the power of Physics-Informed Deep Operator Networks (DeepONet) in the optimization of complex systems, specifically the topology and porosity distribution of adsorption desalination beds. DeepONet is a machine learning framework that combines deep neural networks with physical laws. The model was able to accurately reproduce key physical quantities in the desalination process, specifically uptake, temperature, and pressure, with errors of just 3.7%, 0.26%, and 0.63%, respectively, compared to CFD reference data. The model then moved into the optimization phase, where it applied a particle swarm optimization (PSO) algorithm to search for the optimal configuration of bed topology and porosity. This optimization process resulted in an 18.2% improvement in total water production [Li25].

The findings of these studies are summarized in Table 4.7, which compares inverse approaches across domains, methods, accuracy levels, and computational efficiency.

Table 4.7: Comparison of inverse approaches across domains and methods

Reference	Domain	Method	Key Results
[Cho22]	Hypersonic turbulence calibration	POD + ML surrogate+ Bayesian inference	< 3% deviation
[Jan24]	Turbulent flow over porous media	Domain-decomposed PINN with RANS	5.44%(velocity), 18.9% (k), 4-9% hidden fields
[Wan24b]	Rotating detonation combustor	PIRGAN (GAN +PINN+RNN)	<1% detonation wave error
[Pau24]	Cardiovascular CFD	SECRET (GPs/Res-Nets + Bayesian inference)	0.9-7% (mean), 35-570%(variance)
[Li25]	Adsorption desalination	Physics-Informed DeepONet + PSO	3.7% (uptake), 0.26%(T), 0.63%(p); +18.2% uptake

4.2.5 Design Optimization

This chapter is structured as follows. First, the most commonly used surrogate and optimization algorithms within surrogate-based design optimization are presented. Next, approaches that do not rely on surrogate models are introduced. Then, the engineering domains where AI- and ML-based design optimization is applied are presented. Finally, the validation of the results of the design optimization is covered.

Surrogate Models

A surrogate model predicts the performance of a component without the need for computationally expensive CFD simulations. In aerodynamic shape optimization, for example, it estimates the lift and drag coefficients of unseen airfoils. Figure 4.2 illustrates the most commonly used algorithms for this task.

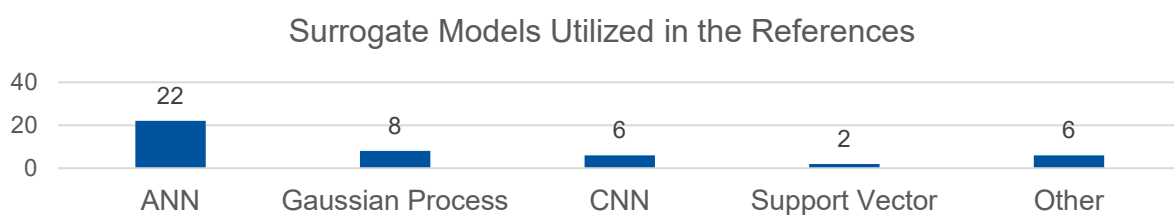


Figure 4.2: Most often used surrogate models for design optimization

The most frequently used surrogate model is based on ANNs. These models are trained either solely on high-fidelity data or on a combination of high- and low-fidelity data. High-fidelity data are obtained from CFD simulations, while low-fidelity data are derived from domain-specific methods that approximate performance characteristics at lower computational cost [Du25].

The second most common approach is the Gaussian Process (GP). This probabilistic model predicts the performance of unseen geometries while also quantifying prediction uncertainty [Taj23].

The third most widely used category are CNNs, a subset of ANNs designed for computer vision tasks. In design optimization, CNNs use visualizations from CFD simulations to predict the performance of designs not previously simulated [Li24].

Optimization algorithms

In general, optimizers are used to find the optimum of a function. In the case of surrogate-based design optimization, the surrogate model acts as the function. Figure 4.3 shows the most commonly used optimizers.

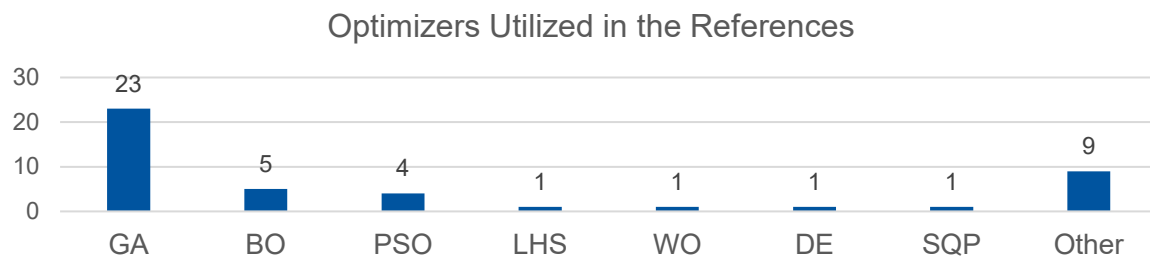


Figure 4.3: Most often used optimization algorithms for design optimizations

Genetic algorithms (GAs) are global optimization methods that mimic the genetic and evolutionary processes of natural organisms. They iteratively improve solution quality through selection, crossover, and mutation within a population of candidate solutions, making them effective for exploring complex design spaces [Xu24a]. In practice, the NSGA-II algorithm (Non-dominated Sorting Genetic Algorithm II) is frequently used.

Bayesian Optimization (BO) is the second most commonly used optimizer. It is designed for problems where evaluating each solution is expensive, such as CFD simulations. It uses a surrogate model, often a Gaussian process, to approximate the unknown function and an acquisition function to decide where to sample next. This balances exploration and exploitation, reducing costly evaluations while efficiently approaching optimal solutions [Zha24d].

The third most commonly used optimizer is Particle Swarm Optimization (PSO), a population-based heuristic inspired by the social behavior of birds and fish. In PSO, candidate solutions, called particles, explore the design space by updating their positions based on individual and collective experience to efficiently identify optimal solutions [Du24].

Non-surrogate-based approaches

DRL methods are most commonly used for shape optimization without surrogate models. These methods use an agent that performs actions and receives rewards or penalties based on their outcomes, with the goal of maximizing the cumulative future reward. In design optimization, an action corresponds to deforming the component, and the agent is rewarded when the deformation improves performance.

For example, NEELAKANDAN ET AL. use DRL to explore the design space of a rocket nozzle, achieving designs superior to those generated by traditional optimization algorithms such as GAs or PSOs [Nee25]. CHEN ET AL. apply DRL to unsteady flow problems, where surrogate models are difficult to construct. They use it for aerodynamic

shape optimization of airfoils to minimize shock wave effects [Che23]. Similarly, LEE ET AL. [Lee24] present a framework that directly integrates DRL with CFD and mesh deformation for multi-objective shape optimization, bypassing surrogate models entirely. Using Proximal Policy Optimization (PPO) with action clipping and reward shaping, their approach maximizes thermohydraulic performance while maintaining computational efficiency through RBF-based mesh morphing and FFD. Their method achieves substantial improvements in Nusselt number, pressure drop, and overall performance, demonstrating the capability of DRL to autonomously learn complex shape-performance tradeoffs over tens of thousands of CFD evaluations.

Domains

AI- and ML-based design optimization is applied across multiple engineering domains and technical components. As shown in Figure 4.4, airfoil optimization is the most common application. In this case, the lift-to-drag ratio of a 2D airfoil is typically maximized under geometric constraints. For example, DU ET AL. train a neural network on both high-fidelity CFD data and low-fidelity panel method data to predict aerodynamic coefficients for unseen airfoil shapes. A particle swarm optimization algorithm is then used to identify the airfoil with the highest lift-to-drag ratio [Du25].

The second most common application is the optimization of heat exchanger geometry. Here, the goal is to find an optimal trade-off between maximizing heat transfer and minimizing pressure drop, often represented by a Pareto front. HE ET AL. train an ANN as a surrogate model to predict the thermal-hydraulic performance of unseen geometries. A multi-objective genetic algorithm is then applied to determine the optimal design [He25].

The third most frequent domain is turbomachinery, where the geometry of compressor or turbine blades is optimized. CHEN ET AL., for instance, increase the pressure ratio and efficiency of a centrifugal compressor using a deep neural network as the surrogate model and a genetic algorithm for aerodynamic blade shape optimization [Che25b].

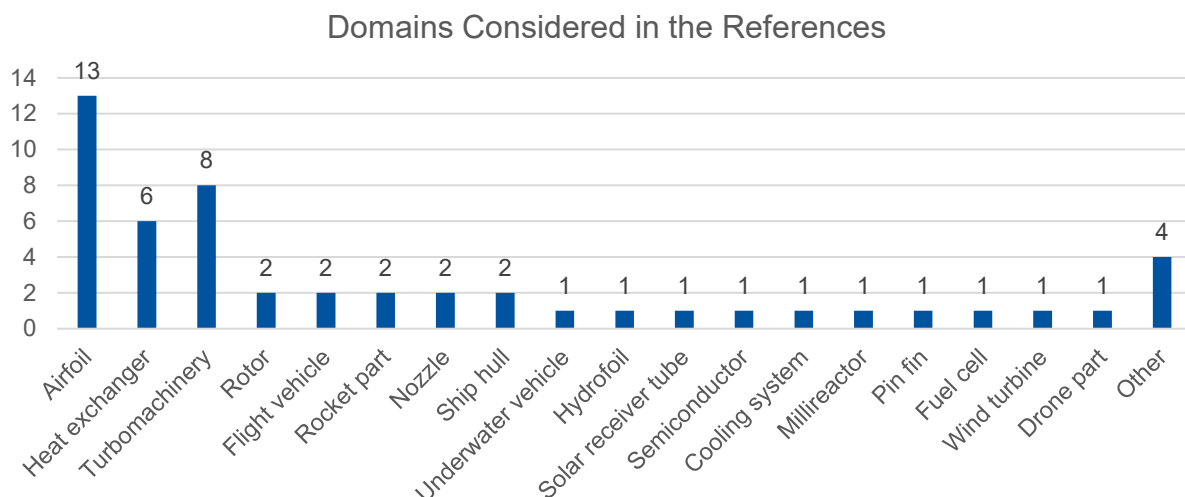


Figure 4.4: Most often considered engineering domains for design optimization

Validation

The result of the design optimization process is the final geometry of the component under consideration. This design can be evaluated either through CFD simulation or by manufacturing and testing the component experimentally.

In most cases, only CFD simulations are performed, as shown in Figure 4.5. Some studies also include experimental validation. For example, SUN ET AL. evaluate their optimized turbine blade structure using both CFD simulations and tests on a 3D-printed prototype [Sun25].

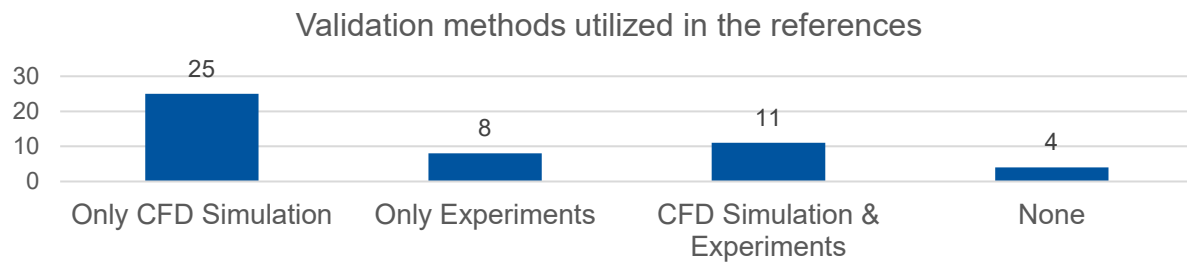


Figure 4.5: Most often used validation methods for design optimization

4.3 Qualitative Analysis

4.3.1 Turbulence Modeling

The qualitative analysis of the turbulence modeling references is summarized in Table 4.8.

Table 4.8: Qualitative Analysis (Turbulence Modeling)

Reference	Novelty	Generaliza- bility	Robustness	Physical Basis	Computa- tional Effi- ciency	Valida- tion	Integra- tion/Auto- mation
[Zhu19]	4	1	3	3	4	2	2
[Nov20]	4	1	3	3	3	2	3
[Sto21]	3	2	1	3	1	3	4
[Lee23]	4	3	4	3	4	3	3
[Zha24c]	3	3	4	3	3	4	4
[Fox24]	4	2	3	4	4	2	2
[Di24]	3	2	3	3	2	3	3
[Fan25]	3	3	3	2	4	2	1
Average	3.5	2.125	3	3	3.125	2.625	2.75

Novelty of Methods

The criterium with the largest average score is “novelty” with 3.5 out of 4. Most novel contributions feature the work by ZHU ET. AL. [Zhu19], who fully replaces traditional turbulence models with machine learning to target high-Reynolds number airfoil flows; NOVATI ET. AL. [Nov20], who uses of RL to derive SGS closure models in LES; LEE ET. AL. [Lee23], who designs a conceptually new ANN-based wall model using input features derived from the FIK identity; and FOX ET. AL. [Fox24], who proposes a closure model using PINNs. Especially FOX ET. AL. demonstrate that accurate solutions can be obtained even from reduced-order or under-constrained formulations, such as enforcing only continuity or partial PDEs as in the Continuity-Only Model (COM). COM achieves good accuracy and compute time (circa 0.74 hours) for velocity prediction using only the continuity equation.

Generalizability

Generalizability poses a challenge for data-based turbulence modeling, which can be seen from its score of 2.125 (smallest score for a criterium here). For example, in the work of ZHU ET. AL. [Zhu19], the model is limited to 2D airfoils in attached, subsonic flows, with training based solely on SA-RANS synthetic data and requiring zone-specific training and manual feature engineering. While there are positive signs for generalizability, such as across different Reynolds numbers and unseen flow/grid conditions, the test cases of the work by NOVATI ET. AL. [Nov20] is limited to forced isotropic turbulence in a periodic cube, and the model does not generalize across discretization schemes without retraining. However, e.g. the approach of ZHAO ET. AL. [Zha24c]:

generalizes across urban layouts and even 2D to 3D scenarios. Though, performance degrades when encountering unseen flow regimes.

Robustness

In robustness, all the references have a rating of larger or equal three, with the exception of the work by STOFFER ET. AL. [Sto21]. There, Robustness has the rating one due to instability during a posteriori testing, where, according to the authors, TKE accumulated at small scales due to error feedback and aliasing. Validation is limited to a single canonical DNS dataset. Two works score a perfect four ([Lee23] and [Zha24c]), and although robustness is good in terms of numerical stability, the rating of [Zhu19] is confined by the limited generalizability confines to three. In the work of FAN ET. AL. [Fan25], the NeuralPDE-Corr model remains stable on coarse grids; the NeuralPDE-Discretize variant, however, fails under some unseen ICs due to flux misprediction and there is no real-time stabilization.

Physical Basis

The highest rating for physical basis belongs to FOX ET. AL. [Fox24]: the PINNs used in this work are grounded in first-principles physics, specifically the 2D incompressible RANS equations. In contrast, the approach of FAN ET. AL. [Fan25] is fully data-driven with no physics-informed constraints or tensor bases, relying on local flow quantities as inputs to a black-box model. The rest of the papers include physically meaningful features, or is grounded in POPE's [Pop75] tensor-basis decomposition for Reynolds stress anisotropy [Zha24c; Di24].

Computational Efficiency

The average score for computational efficiency is 3.125. However, there is a high variance of ratings, ranging from one to four. The ANN of STOFFER ET. AL. [Sto21] incurs a high inference cost, even with additional optimization: about 15 times slower than Smagorinsky SGS, thus deserving a rating of one. On the other side of the spectrum; ZHU ET. AL. [Zhu19], LEE ET. AL. [Lee23], FOX ET. AL. [Fox24] and FAN ET. AL. [Fan25] get a score of four. For example, ZHU ET AL. achieve a speed-up of 35-60% over SA-based RANS and FAN ET. AL. a 26 times speedup compared to DNS (with the hybrid neural solver).

Validation

There is a positive correlation (correlation coefficient = 0.546) between generalizability and validation. The reason for this is that the models that are not validated against a wide range of flows are not shown to be generalizable to them. For instance, the approach of NOVATI ET. AL. [Nov20] validation on complex geometries or boundary layers, thus scoring two in validation as well as one in generalizability. In contrast, the work of ZHAO ET. AL. [Zha24c] is validated on DNS, LES and wind tunnel data across three real-world urban cases, thus getting four in validation, and three in generalizability.

Integration and Automation in CFD-Workflow

Integrationwise, two references score a rating of four: the papers of STOFFER ET. AL. [Sto21] and ZHAO ET. AL. [Zha24c]. STOFFER ET. AL. integrate the ANN SGS model into the LES solver MicroHH and use it during runtime simulations. The model of ZHAO ET AL. is fully embedded into OpenFOAM (simpleFoam), replacing RANS closures via runtime AI predictions. Other references pose several integration challenges such as hand-crafted feature selection ([Zhu19]), manual specification of reward functions ([Nov20]), and lacking automated handling of flow or boundary conditions ([Di24]). The work by FAN ET. AL. [Fan25] score only one, because the prediction is performed offline without any integration into a CFD solver.

4.3.2 Meshing

The results of the qualitative analysis for all included meshing papers are presented in Table 4.9. A score of one (lowest) to four (highest) is assigned to each paper for each criterion. The criterion ‘Mesh Improvement over Baseline’ is only applied to methods for mesh smoothing and adaptation. A blank cell in the table indicates that the authors do not give any information about the respective criterion.

The key findings of the qualitative analysis are presented for each criterion:

Novelty of methods

The only paper scoring a four is the diffusion-based mesh adaptation method by YU ET AL [Yu25]. Because diffusion is considered a state-of-the-art ML technique that outperforms classical approaches, it receives the highest possible score. All other papers score a two or three, indicating the use of a classical ML architecture for the meshing process.

Generalizability

No paper is assigned a score of four, since no meshing method applicable to all practical use cases is proposed. Papers scoring two evaluate their methods on a limited set of similar cases, such as airfoils. Papers scoring three test their methods on a more diverse set of geometries. These may include combinations of airfoils and basic geometric shapes, indicating improved generalization to different shape types, such as it is done by YU ET AL. for their mesh adaptation algorithm [Yu23]. PELISSIER ET AL. evaluate their model across two engineering domains. They apply the algorithm to a deforming plate in structural mechanics and to flow around cylinders in fluid dynamics [Pel24].

Mesh Quality

A score of two indicates a quality comparable to traditional methods. With an average score of 2.5, ML-based meshing methods show slight quality improvements over classical approaches. Seven of the twelve papers reporting mesh quality achieve better results than traditional methods and are assigned a score of three. The mesh adaptation method by MIAO ET AL. is the only exception, as it fails to surpass traditional quality and is therefore scored one [Mia23]. For example, two models show superior

performance. DRL-based mesh adaptation captures physical phenomena with fewer nodes, reducing CFD simulation costs while increasing the accuracy [Fou23]. LIU ET AL. report a 10-30% improvement in mesh quality metrics for their mesh generation model compared to traditional methods [Liu24c].

Table 4.9: Qualitative Analysis (Meshing)

Reference	Novelty of methods	Generalizability	Mesh Quality	Computational efficiency	Integration and Automation in CFD Workflow	Mesh Improvement over Baseline	Validation
[Lim24]	2	2			4		2
[Liu24c]	3	3	3	3	3		2
[Loc23]	3	3	2		4		2
[LU22]	2	3	3		3		3
[Pan23a]	3	3	2		4		2
[Pel24]	2	2		4	4		2
[San24]	2	2	1	3	3		2
[Fou23]	3	2	3	4	2	4	2
[Mia23]	3	2	3	1	2	2	2
[Obi23]	3	3	2	3	2	3	2
[Wan24a]	2	2	3	4	2	3	3
[Wan25a]	2	2	2	4	2	3	2
[WU22]	2	3			2	4	3
[Yu25]	4	3	3	3	3	4	3
[Yu23]	3	3	3	2	3	3	3
Average	2.60	2.53	2.50	3.10	2.87	3.25	2.40

This criterion has the second-highest average score, highlighting the efficiency of ML-based meshing algorithms. Only two papers score one or two, indicating slower or comparable speed to traditional methods. These are the ANN-based mesh generation method by SANCHEZ-GAMERO ET AL. [San24]. and the U-NET-based mesh refinement algorithm by MIAO ET AL. [Mia23]. All other papers reporting efficiency achieve high scores of three or four. ML-based mesh smoothing for example achieves a reported speedup of 10 to 15 times compared to optimization-based smoothing [Wan24a; Wan25a]. The mesh adaptation model by PELISSIER ET AL. reaches a speedup of 63.4 times over CFD-based iterative AMR methods [Pel24].

Integration and Automation in CFD-Workflow

Mesh generation methods automate a larger portion of the CFD pipeline, resulting in a higher average score than mesh improvement algorithms, which require an initial mesh. This initial mesh often needs human intervention, reducing their score. The average score is 3.57 for mesh generation methods and 2.25 for mesh improvement

algorithms. For example, the mesh generation algorithm by PAN ET AL. produces meshes of comparable quality to traditional methods while eliminating the need for human intervention [Pan23a].

Mesh Improvement over Baseline

Almost all mesh improvement methods show significant enhancement over the initial mesh, earning a score of three or four and the highest average rating among all criteria. The only exception is the mesh refinement algorithm based on a U-Net, which fails to achieve a high score [Mia23]. Promising physics-aware mesh adaptation methods have been proposed by WU ET AL. [WU22] and YU ET AL. [Yu23; Yu25]. WU ET AL. use NNs, while YU ET AL. use an MLP and diffusion models to identify regions with flow features, that require a higher mesh density.

Validation

There is a positive correlation (correlation coefficient = 0.38) between the scores for generalizability and validation. Most papers validate their meshing algorithms against CFD results computed on high-resolution meshes. Successful validation across multiple scenarios demonstrates higher generalizability. An exception is the mesh generation algorithm proposed by LU ET AL., where CFD results from the generated mesh are validated against subsonic and transonic wind tunnel data for airfoils [LU22].

4.3.3 Chemical and Combustion Modeling

The results of the qualitative analysis for all included chemical and combustion modeling papers are presented in Table 4.10. A score of one (lowest) to four (highest) is assigned to each paper for each criterion. See Section 3.4.1 for a description of the criteria.

Novelty of methods

Most of the reviewed studies score between two and three in novelty, reflecting the fact that many approaches adapt existing machine learning algorithms to chemical or combustion modeling problems rather than introducing entirely new paradigms. High novelty is found in physics-informed or hybrid surrogate modeling approaches. For example, CHEN ET AL. [Che25a] receive the highest score of four for their dual-loss Physics-Informed Neural Network (PINN) implementation of the WSGG model, which enforces physical consistency during training. CAI ET AL. [Cai25] also achieve a high novelty score due to their hybrid flamelet equilibrium/random data approach combined with MLP-based thermochemical surrogates.

Table 4.10: Qualitative Analysis (Chemical and Combustion Modeling)

Reference	Novelty	Generaliza- bility	Chemical fidel- ity in AI-assisted modeling	Computational Efficiency	Validation	Integration/Au- tomation
[Men21]	3	3	2	4	4	2
[Jai21]	2	2	2	3	2	3
[Xue23]	3	2	2	4	3	3
[Zha25b]	2	2	2	4	3	2
[Che25a]	4	3	2	3	2	4
[Cai25]	3	3	4	3	4	4
[Wan25c]	3	2	3	4	3	3
Average	2.857	2.429	2.429	3.571	3	3

Generalizability

Generalizability scores are mostly moderate (average 2.43), since no method achieves perfect generalization across combustion regimes. Methods trained on highly specific configurations (e.g., HBNG pipeline jet fires [Zha25b], furnace temperature fields [Xue23]) receive a score of two due to their dependence on narrowly defined geometries and operating conditions. In contrast, PINNs [Che25a] and modular MLP surrogates [Cai25] score high because, according to the authors, they can be extended to other fuels, mixture compositions, or solver frameworks with retraining.

Chemical Fidelity in AI-assisted modeling

This criterion shows one of the lowest average scores (2.43), reflecting that most ML-based combustion models do not resolve detailed chemical kinetics directly. Surrogates that emulate thermochemical property tables or ODE solutions, as in CAI ET AL. [Cai25], achieve the highest fidelity score because they preserve the underlying thermodynamics and reaction mechanisms. In contrast, models predicting phenomenological quantities such as jet length or temperature without modeling chemical states [Zha25b; Xue23] are assigned lower scores.

Computational Efficiency

Computational efficiency is a major strength of AI-based combustion modeling, with an average score of 3.57. Surrogate models such as POD+XGBoost [Wan25c] and GDNN-based predictors [Xue23] achieve near real-time inference, enabling online monitoring or digital-twin applications. CAI ET AL. [Cai25] report significant runtime reduction by embedding surrogates directly in the CFD solver, making their approach competitive with reduced-order models while preserving accuracy.

Validation

Validation scores average three, showing a good but not universal commitment to benchmarking against experimental or high-fidelity numerical data. The strongest validation is reported by ZHANG ET AL. [Zha25b], who achieve less than three percent relative error against experimental jet-fire data, and by XUE ET AL. [Xue23], who validate their method with full-scale 350 MW boiler measurements across multiple configurations. Other works such as CHEN ET AL. [Che25a] and MENDIBLE EL AL. [Men21] show good agreement with line-by-line radiation models or experimental RDE data but remain limited to selected operating points.

Integration and automation in the CFD workflow

Integration scores average three, indicating that most methods are partially but not fully automated within CFD solvers. The highest level of integration is achieved by PINNs [Che25a] and MLP surrogates [Cai25], which act as direct drop-in replacements for radiation and thermochemistry modules inside the solver. In contrast, reduced-order models [Men21] and ANN+GA frameworks [Jai21] are primarily used as offline tools for design and analysis, limiting their integration score.

4.3.4 Inverse Problems

Table 4.11 summarizes the qualitative evaluation of all the inverse problem studies included in this review. Each study is rated on a scale of one (lowest) to four (highest) for the defined criteria.

Table 4.11: Qualitative Analysis (Inverse Problems)

Reference	Novelty	Generaliza- bility	Uncertainty Handling	Computational Efficiency	Validation	Integration/Au- tomation
[Pau24]	3	3	3	3	3	3
[Li25]	4	4	4	3	2	4
[Cho22]	3	3	3	3	3	2
[Zhu22]	3	2	2	3	2	2
[Wan24b]	4	3	1	3	2	2
[Jan24]	4	4	2	2	3	2
[Yan25]	2	2	1	3	2	2
[Cai24]	3	2	1	3	1	3
[Jal24]	3	3	2	2	2	3
[Rad24]	3	3	1	3	2	3
Average	3.2	2.9	2	2.8	2.2	2.6

Novelty of Methods

Most studies score a three in this category, reflecting strong innovation in how inverse problems are formulated using machine learning frameworks. Methods introducing PINNs [Jan24] and generative adversarial architectures with recurrent components

(PIRGAN) [Wan24b] are rated four, as they incorporate advanced learning techniques beyond classical surrogate modeling. Operator learning approaches, such as the DeepONet framework applied to inverse design [Li25], also receive the highest score due to their hybrid use of physics, operators, and optimization. Approaches relying on more established techniques, like Gaussian process emulators or CNN-based regressors, are rated two or three depending on whether they adapt or extend existing algorithms.

Generalizability

No method achieves complete generalization, but several demonstrate strong performance across different flow types or geometries. Studies scoring two evaluate inverse models on narrow scenarios with fixed boundary conditions or domain geometries, as in the CNN regression approach by [Zhu22]. Methods scoring three test their models under multiple physical conditions, such as different Reynolds numbers or boundary types, as in [Cho22]. Only two methods receive a score of four: [Jan24], who extrapolate PINNs across flow types using domain decomposition, and [Li25], whose operator network handles changes in topology, porosity, and inlet conditions in a unified framework.

Uncertainty Handling

This remains the weakest area across the dataset, with an average score of just two. Most studies use deterministic ML models that do not include any form of uncertainty quantification. For example [Wan24b], [Yan25] and [Cai24] do not report any posterior estimates or confidence intervals and are scored one. The only two studies that go beyond deterministic output are [Pau24] and [Cho22] each receiving a three for incorporating Bayesian inference or emulator-based posterior sampling. [Li25] is a notable exception as the DeepONet-based inverse design framework integrates probabilistic mapping and ensemble techniques to quantify confidence in predictions.

Computational Efficiency

Inverse methods generally perform well on this criterion, achieving an average score of 2.8. Surrogate-based and CNN-based models demonstrate significant speed advantages once trained, earning scores of three. Conversely, optimization-heavy or physics-rich networks are less efficient due to high runtime or solver coupling and receive lower scores of one and two, respectively.

Validation

The average validation score across all studies is 2.2, indicating that most inverse problem models are moderately tested against real-world or high-fidelity reference data. A score of two is assigned to methods that validate predictions on synthetic datasets or limited numerical cases. Methods scoring three validate their inverse solutions in multiple settings or with experimental input, as seen in [Pau24], while [Yan25] test against aeroelastic response data from wind tunnel experiments. No study receives a score of four, as none demonstrate fully generalized, real-world validation across multiple systems or domains.

Integration and Automation in CFD workflows

The degree to which inverse methods are embedded in CFD workflows varies, with an average score of 2.6 across studies. Some approaches are explicitly designed for deployment and can be incorporated directly into simulation or optimization loops. [Li25] receives the highest score for its fully automated inverse design pipeline, where DeepONet is coupled with a particle swarm optimizer. [Cai24] and [Rad24] also achieve high scores, as their models operate within broader simulation workflows and support feedback-based refinement. Several studies score lower due to their limited automation and reliance on pre-processing or external CFD solvers

4.3.5 Design Optimization

The results of the qualitative analysis for design optimization are presented in Table 4.12 ordered by year of publication.

Table 4.12: Qualitative Analysis (Design Optimization) (1)

Reference	Novelty	Generalizability	Design Quality	Computational Efficiency	Validation	Integration/Automation
[Mar23]	3	1	3	4	2	2
[Pid23]	2	2	3	4	2	2
[Pog23]	2	2	3	4	2	2
[Giu23]	3	1	4	4	3	2
[Taj23]	2	2	3	4	3	2
[Hon23]	2	1	3	3	2	2
[Liu23]	3	2	1	3	2	2
[Net23]	2	1	2	1	1	2
[Had23]	2	2	4	1	2	2
[Sik23]	2	2	3	3	1	3
[Aso23]	3	3	3	4	2	3
[Beg23]	3	3	3	2	2	3
[Che23]	3	2	3	2	2	3
[Wan23]	3	2	3	3	2	2
[Ker23]	3	2	3	2	1	3
[Sun23]	3	3	1	2	2	2
[Li24]	3	2	3	4	3	3
[Sha24]	2	2	4	4	4	2
[Xu24a]	3	2	3	4	2	2
[Du24]	3	1	3	3	2	2

Table 4.13: Qualitative Analysis (Design Optimization) (2)

[Zha24a]	2	3	3	1	2	2
[Shu24]	3	3	2	4	2	3
[Kum24]	3	2	3	3	2	2
[Lee24]	4	2	4	3	3	3
[Cho24]	3	2	3	2	2	3
[Liu24b]	3	2	3	2	3	3
[Xu24b]	3	2	3	3	2	2
[Sun24b]	3	2	3	3	2	3
[Zha24b]	3	2	3	3	2	2
[Pad24]	3	2	3	4	2	2
[Jia24]	3	3	3	4	2	2
[Esf24]	4	3	3	2	2	2
[Lei24]	3	3	2	2	2	2
[Zha24d]	2	2	3	3	3	2
[Gho24]	3	3	3	2	3	3
[Du25]	3	2	3	4	1	2
[Wan25b]	2	2	3	1	2	2
[Rav25]	3	2	3	4	2	3
[Par25]	2	2	3	3	3	3
[Saw25]	2	2	3	3	3	2
[Liu25]	3	2	2	2	1	1
[Zha25c]	3	2	1	2	1	2
[Nee25]	3	1	4	1	1	3
[Sun25]	3	1	3	2	2	2
Average	2.75	2.05	2.89	2.82	2.10	2.32

The key findings and the most promising approaches are presented for each criterion:

Novelty of methods

Regarding novelty, only two studies propose original architectures that could serve as new standards, each receiving the highest score of four. The first is the DRL-based framework by LEE ET AL., which optimizes heat exchanger structures without surrogates and thereby accelerates the optimization process significantly [Lee24]. The second is by ESFAHANIAN ET AL., who apply deep learning combined with surrogate models for aerodynamic shape optimization of gas turbines [Esf24]. In contrast, the common surrogate-based frameworks coupled with optimization algorithms typically score two or three, depending on the sophistication of the methods used.

Generalizability

No study achieves a score higher than three in this criterion. Among the more promising approaches is the aerodynamic shape optimization proposed by ASOUTI ET AL., which employs an RBF surrogate combined with an evolutionary algorithm [Aso23]. Another method scoring three is presented by ZHANG ET AL., who use an NN and RBF as surrogates together with a GA to optimize airfoil geometry under varying cruise conditions [Zha24a].

Design Quality

Five studies achieve a score of four in this criterion, reflecting high practical applicability and the inclusion of key constraints such as structural integrity and manufacturability. The DRL-based rocket nozzle optimization proposed by NEELAKANDAN ET AL. generates superior designs compared to traditional algorithms such as PSO or GA [Nee25]. Surrogate-based methods also prove effective. For example, SHAHRABI ET AL. employ an ANN surrogate with a GA to optimize compressor design, incorporating critical constraints to ensure feasibility and practicality [Sha24].

Computational efficiency

This criterion achieves the second-highest average score, highlighting the efficiency of ML- and AI-based design optimization. SHUKLA ET AL. report a speedup of approximately 32,000 times in predicting flow fields around an airfoil compared to CFD simulations using deep neural operators [Shu24]. Similarly, SUN ET AL. demonstrate speedups of several orders of magnitude in their surrogate-based airfoil optimization framework [Sun24a].

Validation

In most cases, the optimized design is validated through CFD simulations. In some studies, the component is additionally manufactured and tested experimentally. One notable example is the work of SHAHRABI ET AL., who achieve a score of four by extensively testing their optimized gas turbine under multiple operating conditions in an industrial setting [Sha24].

Integration and Automation in CFD-Workflow

Almost all papers score a two or three for this criterion, indicating that manual effort remains in the design optimization workflow. Typical examples include preparing CFD data for training surrogate models and performing validation, which is still largely carried out manually.

4.4 Research Gaps

The studies in this systematic literature review explicitly state research gaps, as identified through the search strategy outlined in Chapter 3, where search strings are designed to capture publications that discuss gaps and potential future research directions after presenting their findings. These identified gaps provide insights that directly contribute to addressing both the first and second research questions.

4.4.1 Turbulence Modeling

In the reviewed references for turbulence modeling, several research gaps are mentioned by more than one paper; with generalization and validation concerns being the most typical one.

Generalization and Validation Concerns

All eight references mention lacking validation and generalization as a concern. Among them: ZHU ET AL. [Zhu19] state that their current work is limited to attached flows, where the boundary layer remains connected to the surface, and highlight the need to extend their framework to more complex scenarios, including separated flows and shock-influenced flows. NOVATI ET AL. [Nov20] recognize that their current framework is tested on forced isotropic turbulence, which, while useful for proof-of-concept, lacks realism compared to practical flows. STOFFER ET. AL. [Sto21] suggest CNNs (like the one proposed by ZHAO ET. AL. [Zha24c]) to improve spatial generalization and reduce computational cost. ZHAO ET AL. [Zha24c] recommend extending the current RANS framework to a URANS (Unsteady RANS) to better capture transient flow behavior, as well as the expansion and diversification of the training dataset. The authors observe that model performance tends to degrade in underrepresented regions. The present work of FAN ET AL. [Fan25] is restricted to 2D Kolmogorov turbulence. According to the authors, this prompts a need for extension to fully 3D turbulent flows. They also highlight the importance of improving solver stability on coarse and unstructured grids, as current results are demonstrated on structured meshes.

A-priori Instability and Deployability

According to STOFFER ET AL. [Sto21], a major limitation of their proposed ANN-based SGS model is the a posteriori instability of the model. Despite showing strong performance in a priori tests, the ANN leads to error accumulation and aliasing when used in actual LES runs. The authors recommend improving the training process by incorporating online or dynamically coupled learning strategies, such as co-training with DNS or using adaptive methods like those proposed by RASP (2020) [Ras20] and GUAN ET. AL. (2021) [Gua21], where the training of the ANN SGS model would be within the online LES simulation. Another way would be to add the expected errors introduced by the a posteriori simulation to the filtered u , v , w during training.

Lack of Physical Constraints

From a physical modeling standpoint, STOFFER ET. AL. [Sto21] acknowledge that their current ANN lacks physical constraints, and the authors advocate embedding conservation laws, realizability conditions, or other physics-informed priors directly into the architecture. ZHAO ET AL. [Zha24c] emphasize the need for improved modeling of the turbulent kinetic energy discrepancy (Δlnk), which currently lacks physical constraints and can result in nonphysical predictions, particularly in complex or highly anisotropic flow regions. FOX ET AL. [Fox24] call for deeper investigation into how simplistic turbulence assumptions, such as the mixing length model, affect solution quality and

convergence behavior, particularly in flows where such assumptions may not hold. FOX ET AL. [Fox24] address this problem by utilizing PINNs.

4.4.2 Meshing

The papers highlight shortcomings of their methods and propose directions for future work. The most common research gaps are the restriction to 2D geometries and the dependency on training data.

Simple ML architectures

Several papers indicate plans to improve their methods by exploring more advanced or alternative ML architectures. LU ET AL. propose using DRL or deep learning for their mesh generation algorithm [LU22]. LIU ET AL. suggest testing neural network architectures beyond the one used in their method [Liu24c]. PELISSIER ET AL. recommend exploring alternatives to the GNN architecture applied in their mesh generation and adaptation approach [Pel24].

Limitation to 2D cases

A second limitation is that all but one paper restrict their methods to 2D geometries. Five papers propose extending their approaches to 3D, including methods for mesh generation [LU22], adaptation [Mia23; WU22], and smoothing [Wan24a; Wan25a].

Limited inclusion of physical information

PELISSIER ET AL. acknowledge the absence of physical information in the mesh generation and adaptation process. Incorporating physical constraints could reduce the amount of required training data [Pel24].

Dependency on training data

YU ET AL. acknowledge the limited availability of training data for diverse geometries, therefore restricting their mesh adaptation to airfoil and missile cases [Yu25]. WANG ET AL. propose using more complex training meshes to enhance the robustness of their mesh smoothing algorithm [Wan24a].

4.4.3 Chemical and Combustion Modeling

While the reviewed studies demonstrate promising advances in accelerating CFD and combustion modeling with AI, several overarching limitations occur across the literature, pointing to where future research can deliver the greatest impact.

Limited Parameter Space and Generalization Constraints

Many models are trained on CFD datasets that cover only a restricted range of operating conditions, such as nozzle diameters, pressures, or blending ratios. For example, ZHANG ET AL. [Zha25b] limit their surrogate to hydrogen blending of 0-30% and pressures of 2-12 MPa, while XUE ET AL. [Xue23] observe reduced predictive accuracy when conditions deviate from the training set. Such scope constraints limit generalization to new fuels, geometries, or boundary conditions. Future work is therefore aimed at expanding training sample diversity and extending applicability to higher pressures, broader nozzle ranges, and different combustion configurations [Zha25b; Xue23].

Chemical Fidelity and Physics Integration

A recurring gap is the simplified treatment of combustion chemistry. JAIN ET AL. [Jai21] explicitly acknowledge that reduced Primary Reference Fuel (PRF) and NO_x mechanisms cause discrepancies in emission predictions, while CAI ET AL. [Cai25] still assume inert nitrogen despite otherwise detailed kinetics. To address this, future research emphasizes incorporating more detailed kinetic mechanisms, refining NO_x modeling (including low-temperature chemistry), and improving the integration of real-gas thermodynamics with surrogate models [Jai21; Cai25].

Validation and Robustness

Most works validate models within narrow ranges of their original parameter space. For instance, CHEN ET AL. [Che25a] compare their PINN-WSGG framework only against spectral benchmarks, but not under varying gas compositions or pressures. XUE ET AL. [Xue23] also report that their correction framework is less effective in furnace zones distant from sensors. Future directions highlight the need for broader experimental validation campaigns, extension of frameworks to new gas mixtures and pressures, and refinement of correction strategies across diverse flow regions [Che25a; Xue23].

Uncertainty Quantification

Few studies include formal frameworks for quantifying predictive uncertainty. While ZHANG ET AL. [Zha25b] perform sensitivity checks, they stop short of uncertainty propagation, and WANG ET AL. [Wan25c] note reduced robustness under limited training data. Future work emphasizes the implementation of explicit uncertainty quantification frameworks, improved robustness under sparse data regimes, and the application of adaptive sampling for uncertainty-aware surrogate modeling [Zha25a, Wan25c].

4.4.4 Inverse Problems

The reviewed studies also acknowledge several limitations and outline directions for future research. The most frequently cited challenges include the lack of uncertainty quantification, limited generalization across domains, and weak integration into real CFD workflows.

Deterministic models without uncertainty quantification

Several methods produce point predictions without estimating uncertainty or confidence intervals. WANG AND WEN ET AL note the deterministic nature of their PIRGAN model, which lacks probabilistic output [Wan24b]. JANG and ZHU report similar limitations in their PINN and CNN-based approaches, respectively [Jan24; Zhu22]. Only PAUN and CHOWDHARY incorporate Bayesian inference, yet their methods show considerable posterior variance and poor robustness under noisy conditions. [Pau24; Cho22].

High dependency on CFD training data

Many learning-based inverse methods require large high-fidelity datasets generated through forward simulations. PAUN explicitly mention the computational cost of

producing CFD training data for the SECRET emulator [Pau24]. CHOWDHARY face similar constraints when constructing surrogate models for his hypersonic flow calibration [Cho22]. Reducing this dependency via unsupervised learning or physics-informed training can be a potential future research direction.

Manual integration into solver pipelines

Several inverse models are not directly integrated into CFD workflows and require manual configuration. WANG and JANG both reconstruct high-resolution fields but leave solver integration to the user [Wan24b; Jan24]. ZHU similarly predict parameters without automating their application in simulation loops [Zhu22]. Future work can focus on automating the integration of inverse models into CFD pipelines, enabling end-to-end workflows that apply predictions as boundary conditions or design modifications.

4.4.5 Design Optimization

Research on the integration of AI and ML with CFD-based design optimization reveals several critical gaps. Current methods often lack generalizability, sufficient integration with industrial workflows, and robust handling of data and physical complexity. Additional challenges include incorporating multi-objective constraints, advancing model architectures, and addressing the high computational and manufacturing costs of optimized designs.

Generalizability and Validation Concerns

Proposed design optimization frameworks are typically applied to a single component and validated only under one set of operating conditions, which limits their generalizability. For example, WEN ET AL. optimize a specific HVAC structure without validating performance under varying conditions such as room size, flow rate, or ambient temperature [Wen25]. NETTO ET AL. acknowledge that their optimization of wind turbine blades does not consider varying operating conditions [Net23]. LIU ET AL. apply their method exclusively to bulbous bow optimization, without extending it to full-hull or more complex geometrical modifications such as stern or midship sections, making broader geometric design a natural next step [Liu24a]. PIDAPARTHI ET AL. identify experimental testing of additively manufactured optimized designs as a long-term goal, but their current work lacks physical validation [Pid23]. LEE ET AL. [Lee24] demonstrate strong thermohydraulic performance improvements but apply their framework only to 2D micro-channel cases. WAN ET AL. [Wan23] validate primarily against a single literature source, and RAV ET AL. [Rav25] restrict their study to 2D single-pin cases at low Reynolds numbers. BEG ET AL. [Beg23] present only grid convergence checks, without experimental comparison.

Integration and Automation

Current design optimization methods are not fully integrated into CFD pipelines or industrial tools, which limits their practical applicability. XU ET AL. highlight the need for adaptive optimization algorithms with dynamic retraining, tighter integration into industrial CFD/CAD workflows, and automatic hyperparameter tuning [Xu24a]. WEN ET AL. point out that their framework is static and offline, lacking real-time coupling or adaptivity, and suggest advancing the pipeline through active learning, adaptive mesh refinement, or solver-in-the-loop optimization [Wen25]. HONG ET AL. emphasize the development of adaptive, real-time CFD–AI integration, for example in applications such as in-flight rotor geometry adjustment [Hon23]. Even frameworks that bypass surrogate models face integration challenges: LEE ET AL. [Lee24] and PADULA ET AL. [Pad24] rely on RBF mesh morphing to avoid costly remeshing, but mesh failures and quality checks still require manual intervention. SAWANT ET AL. [Saw25] train their RL agent using a low-fidelity panel method but do not describe a closed-loop retraining mechanism with CFD, leaving final designs dependent on separate validation.

Data and Model Fidelity

The quality of design optimization strongly depends on the amount and quality of training data available for surrogate models. XU ET AL. emphasize the need to expand dataset size to improve model performance [Xu24a]. JIANG ET AL. identify prediction errors in certain regions of the design space as a result of insufficient data density and suggest improved sampling strategies, such as adaptive sampling, to ensure better coverage and reduce uncertainty [Jia25]. WANG ET AL. [Wan23], LIU ET AL. [Liu24b], and PARK ET AL. [Par25] highlight the risk of surrogate extrapolation outside training data, degrading design reliability. PADULA ET AL. [Pad24] find that limited training data combined with PCA projection can bias shape representations and even drag predictions. ASOUTI ET AL. [Aso23] note that sampling strategy strongly affects surrogate quality, and coarse clustering can harm boundary fidelity.

Simplified Assumptions and Missing Physics

Simplified assumptions in CFD data remain a fundamental limitation of surrogate models. While such approximations enable computationally efficient simulations over large design spaces, they often come at the expense of physical fidelity. WEN ET AL. assume constant wall temperature and uniform airflow, noting that while such simplifications are common, they reduce real-world applicability [Wen25]. LI ET AL. exclude geometric constraints such as thickness and curvature, which are essential for structural strength and manufacturability [Li24]. JIANG ET AL. focus solely on aerodynamic metrics such as lift, drag, and center of pressure, and highlight the need to incorporate aeroelasticity, structural weight, and thermal loads for more realistic multi-physics optimization [Jia25]. SAWANT ET AL. [Saw25] train RL agents with inviscid, incompressible panel methods that neglect viscous and separation effects, potentially biasing shape selection, while CHOI ET AL. [Cho24] assume perfectly insulated walls and neglect natural convection effects.

Multi-objective Optimization

Omitting real-world constraints, such as manufacturing feasibility, can reduce the applicability of optimization results. WEN ET AL. emphasize the importance of multi-objective optimization that includes factors such as cost, weight, or noise in addition to thermal performance [Wen25]. TAJ ET AL. propose extending optimization to additional design variables, including flight control surfaces, deflection angles, and performance metrics such as endurance or take-off distance [Taj23]. LI ET AL. suggest the joint optimization of hydrofoil profile, chord length, and twist angle to improve tidal turbine performance [Li24]. CHOI ET AL. [Cho24], WAN ET AL. [Wan23], and SAWANT ET AL. [Saw25] include only minimal thickness or weight constraints, ignoring manufacturability or structural robustness. XU ET AL. [Xu24b] and PARK ET AL. [Par25] omit thermal, structural, and fabrication constraints entirely, while PADULA ET AL. [Pad24] support only linear constraints and introduce discretization error after mesh morphing.

Model Improvements

Several studies highlight the need for more advanced ML architectures to improve design quality. TAJ ET AL. stress the importance of scalability, particularly in adapting methods to larger and higher-dimensional design spaces [Taj23]. NEELAKANDAN ET AL. propose enhancements to CNN and DRL models for their method for rocket nozzle shape optimization [Nee25]. SHAHRABI ET AL. recommend physics-informed or multi-fidelity surrogate models, noting that their ANN was trained only on mean-line data, which limits prediction quality and increases dependence on empirical calibration [Sha24].

High Computational Cost and Scalability

High costs arise in two main ways: improved surrogate models require larger and more complex training datasets, which increase the number and cost of CFD simulations, while optimized designs may be more difficult and expensive to manufacture. GIUFFRÉ ET AL. note that their model does not simulate the full compressor assembly in CFD due to computational expense and limited benefit in this configuration [Giu23]. CHEN ET AL. highlight that the use of free-form surface geometries can significantly raise manufacturing costs and complexity [Che25b]. More recent DRL- and Bayesian Optimization-based frameworks still face scalability limits. LEE ET AL. [Lee24] report that per-pin individual deformation requires $\approx 23,000$ training episodes and substantial GPU + CPU parallelism. CHEN ET AL. [Che23] note that buffet optimization is still CFD-dominated, XU ET AL. [Xu24b] report ≈ 50 minutes per HF evaluation and BEGALL ET AL. [Beg23] require ≈ 6.5 h per CFD run even after Bayesian optimization.

5 Discussion

This chapter reflects on the methodology and the results presented in the preceding sections. After analyzing the methodology used for the systematic literature review, the research questions are answered, with the first research question for the four different research areas of turbulence modeling, meshing, chemical and combustion modeling and inverse problems dealt with separately.

5.1 Discussion of the Methodology

The inclusion of additional search strings considerably reduces the number of papers identified, which carries the risk that some relevant studies may be excluded if they do not explicitly use these terms in their titles or abstracts but discussed them within the full text. However, this approach makes it possible to focus the analysis more effectively on the body of work that is captured, ensuring a clearer and more concentrated examination.

The chosen evaluation criteria, general and domain-specific, are closely interrelated with each other. Understanding these connections helps interpreting the results of the literature review in a deeper level. For example, validation and generalizability are strongly connected: a method that has been thoroughly validated across various scenarios is more likely to generalize well to new cases, provided the method is tested to have satisfying results in a wide range of applications. This interdependency is particularly relevant when assigning a novelty rating, such that a method can only achieve the highest novelty score if it not only presents an original idea but also demonstrates high performance in critical criteria such as generalizability, computational efficiency, or relevant domain-specific factors like mesh quality or design quality. Moreover, high validation is essential for a novelty rating of 4, as scientific rigor requires that new methods be demonstrably reliable and effective, not just innovative. Moreover, in turbulence modeling, robustness is often directly tied to generalizability, since a turbulence model that handles a wide range of numerical conditions without failure inherently reflects broader applicability.

5.2 Discussion of the Research Questions

RQ1: What are the gaps in the current literature regarding the integration of AI in CFD workflows, and how can future research address these gaps?

Turbulence Modeling

The studies demonstrate a broad range of AI methods for turbulence closure modeling, employing CNNs, ANNs, LSTMs etc., tailored to capture e.g. spatial, temporal, or wall-related effects. Both supervised and reinforcement learning paradigms are used, with multi-agent RL proposed to address the poor generalization of supervised methods. [Nov20] Yet, the approaches remain specialized: e.g. for subsonic attached flows [Zhu19], isotropic turbulence [Nov20], isothermal conditions [Sto21]; or using wall-

stress models unsuitable for separation [Lee23], CNNs for steady flows [Zha24c], and CNN–LSTM hybrids for transients [Fan25]. Many are further limited to incompressible 2D simulations, restricting industrial use. Due to this lack of global generalizability, the models generally achieve low robustness ratings, despite high robustness for the respective specific test cases.

In general, the models achieve high inference speedups compared to baseline methods. However, having to train a model for every distinct use case due to low generalizability hinders the speedup.

Another barrier is the incomplete integration of these models into CFD workflows, since e.g. STOFFER ET. AL. [Sto21] report model instability in a posteriori (online) deployment of the model in a CFD solver. Furthermore, the absence of standardized benchmarking datasets hampers meaningful comparison across methods and make it difficult to assess relative performance objectively.

In essence, AI-based turbulence modeling is still in an exploratory stage. Future research could address the gaps by developing hybrid approaches that combine the strengths of existing architectures and validate them on a substantially larger and more diverse set of test cases. The ultimate ambition would be a globally robust and fully integrated, unified data-driven turbulence model. However, achieving such a model would be a significant challenge, comparable to the long-standing difficulty of unifying different CFD codes into a single universal solver. Just as traditional CFD employs different models for different scenarios, machine-learning-based approaches also appear to be moving toward specialized models with trade-offs. So, validation must remain the central priority to stand a realistic chance of replacing the interpretable and thoroughly tested CFD models, especially in safety-critical applications.

Meshing

Current methods show limited generalizability, often restricted to airfoils or specific geometry test sets, where extensive CFD training data is already present [Yu25]. Some algorithms may be applicable to more diverse and practically relevant geometries, but this capability is not tested or validated. RL-based approaches could address this limitation by reducing or eliminating the need for large training datasets [Fou23]. Moreover, most methods remain limited to 2D geometries, restricting their applicability to real-world 3D problems. Finally, mesh quality often matches and sometimes exceeds that of traditional methods. A reason might be that further optimization might not be possible.

That is, ML-techniques are able to speed up the meshing process and produce meshes of higher quality. To ensure practical usability, the generalization to different shapes and to three dimensional geometries is needed.

Chemical and Combustion Modeling

The reviewed studies show that AI-assisted combustion and chemistry modeling is advancing quickly but remains fragmented, with most methods targeting narrow problems. Surrogates for jet fires [Zha25b], physics-informed radiative transfer [Che25a], and ROMs for detonation dynamics [Men21] offer strong efficiency gains but lack generalization across fuels, operating regimes, and geometries. Authors consistently call for broader training datasets, higher chemical fidelity, and improved generalization via transfer learning or physics-guided extrapolation to make models more portable to industrial systems [Zha25b; Cai25; Xue23; Jai21]. Expanding validation campaigns beyond narrow canonical cases is critical to build trust, especially for safety-critical scenarios. Uncertainty quantification is another gap, as most studies report point estimates without confidence bounds [Wan25c; Cai25]. Finally, deeper integration of surrogates into CFD solvers, co-developed with training and inference pipelines, is necessary to enable seamless design optimization and control.

In summary, AI-driven combustion and chemistry models show strong progress, but their broader impact will hinge on developing scalable, transferable, and uncertainty-aware frameworks that can operate reliably across diverse fuels and operating regimes.

Inverse Problems

Despite recent progress, the integration of artificial intelligence into CFD workflows for inverse problems remains partial and inconsistent. Many models only work as prediction tools, requiring manual post-processing or external solver coupling to apply their outputs within simulation loops [Jan24; Wan24b]. This limits their utility in real-time design, control, or adaptive refinement. Only a few studies, such as those by LI and RADHAKRISHNAN, shows fully connected architectures where AI-driven inverse modeling is used to guide simulations or optimize system configurations directly.

To address these gaps, future research should focus on making it easier to connect AI models directly with CFD solvers, enabling two-way interaction between predictions and simulations, and creating standardized methods for smoothly integrating AI into existing engineering workflows. Additionally, enabling such integration in 3D and multi-physics settings will be key to advancing applicability in real-world scenarios.

RQ2: In what ways do AI techniques enhance the design optimization process within the CFD pipeline, and what challenges arise during their application?

AI techniques significantly enhance CFD-based design optimization by reducing computational cost, accelerating convergence, and enabling exploration of larger design spaces. Surrogate models such as Gaussian Processes and Neural Networks replace expensive CFD evaluations with fast approximations, supporting global search and multi-objective optimization [Wan23]. Reinforcement learning and Bayesian optimization provide adaptive, data-efficient search strategies, while physics-informed networks and multi-fidelity surrogates improve robustness and capture missing physics

[Sha24]. Integration with adaptive sampling and active learning further reduces the number of required high-fidelity simulations, bringing workflows closer to solver-in-the-loop optimization [Xu24a].

However, several challenges limit their industrial adoption. Most methods remain problem-specific, validated only on single components or narrow operating conditions, which restricts generalizability [Liu24b; Net23]. Data scarcity and uneven sampling create surrogate bias and extrapolation errors [Wan23], while simplified physics and missing multi-physics coupling reduce real-world applicability [Cho24]. Integration into CFD pipelines is often incomplete, requiring manual interventions for meshing or parameter tuning [Lee24; Pad24]. Finally, robust uncertainty quantification (UQ) and comprehensive validation are still underdeveloped, leaving decision-makers with limited confidence in AI-generated designs [Beg23].

Overall, AI techniques clearly enhance design efficiency and expands the feasible design space, but widespread deployment will require progress in generalization, automation, UQ, and integration with industrial workflows.

6 Conclusion and Outlook

In summary, the literature shows that AI-assisted CFD is an exploratory field, where there is no single universal method. Surrogate models remain common, while there is growing interest in PINNs and reinforcement learning that aim to solve the problems of pure data-driven supervised methods.

Key gaps remain in generalizability, integration with industrial workflows, and validation. Also, uncertainty quantification and manufacturability constraints are rarely included but are crucial for safe and economically viable deployment.

Future research should prioritize diverse training datasets, broader validation, tighter coupling of AI methods with CFD solvers, and developing transferable models that can support automated design loops.

The methodological decisions adopted in this review directly influence the scope of the results. One central choice is the restriction of the search strings through the inclusion of terms such as “research gaps” and “future research.” While this provides a highly targeted set of studies, it also risks excluding relevant contributions that address limitations and challenges more implicitly. A broader formulation of search strings in future reviews could therefore capture a wider range of perspectives and ensure that subtle but important insights are not overlooked.

Another methodological consideration concerns the screening process. While the selection relies primarily on database outputs, the systematic examination of reference lists from high-impact papers could further enhance coverage. In particular, tracing forward citations would allow the identification of follow-up studies and emerging contributions that expand upon earlier research, thereby providing a more continuous picture of scientific progress.

The scope of the present review is limited to turbulence modeling, meshing, chemical and combustion modeling, inverse problems and design optimization. Future reviews may adopt two complementary strategies: expanding the scope to include additional CFD-AI integration domains, or narrowing the scope to focus on a single research area in depth.

The choice of the cut-off date also influences the scope of the review. For RQ2, the search string is restricted to publications after 2022. While this is suitable for capturing recent developments, an earlier cut-off date could yield a more general perspective, which allows the identification of trends and shifts in research directions over time.

Finally, the adoption of PRISMA as a methodological framework provides a structured foundation and ensures transparency in the review process. Its systematic approach helps in organizing the inclusion and exclusion criteria

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